

Mechanistic Tomography: Designed Measurement for Control-Oriented Interpretability

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Abstract

Understanding and steering transformer behavior requires measuring internal states and intervention effects that the model does not report directly. Coordinate patching, attribution patching, subset interventions, and Hessian-vector products obtain different measurements under different access assumptions. We formulate their shared measurement problem as **mechanistic tomography**: for a declared basis and intervention family, $\tilde{y} = Ax + w$ relates the effect map x to designed measurements, with w accounting for non-linear response, sampling error, and basis misspecification. The common form makes recoverability, error amplification, and measurement cost properties of the design that can be checked, while preserving each method’s target.

Our theory connects measurement error to perturbation scale, calibration dimension, and interaction recovery. It supports a practical sequence: start with the least costly measurement family, check finite responses on held-out interventions, calibrate a simple mismatch, and add interactions only when a structured residual remains. When an estimate guides an intervention, it acts as an observer, so we also test its effect on control.

In a two-HMM wind tunnel, control error tracks observer error under a fixed controller and actuator (Spearman 0.95). Aggregate probes recover a compressible map using 12 measurements instead of 32 coordinate interventions, and eight finite probes calibrate attribution patching to held-out $R^2 = 0.948$ at the larger intervention scale. Planted interactions and compiled transformers show when the feature family or basis must change. On GPT-2-small indirect-object identification, one cross-group interaction reduces held-out mean absolute error (MAE) by about one-third. On Qwen-2.5-7B, a calibrated additive map gives held-out $R^2 = 0.983$ on a finite decline-versus-compliance response surface, with no detected MAE benefit from pairwise lifting. These studies show how held-out responses guide both refinement and stopping.

1 Introduction

Mechanistic interpretability often asks a simple question: how much does an internal component contribute to a model’s output? The model does not report that effect directly, so we have to measure it. Coordinate patching changes one component at a time. Attribution patching

uses gradients to estimate many small local effects. Subset interventions change several components together, while Hessian-vector products probe interactions. These methods use different forms of access and may estimate different quantities, but they face the same basic problem: recovering an internal effect from incomplete measurements.

We formulate that problem as $\tilde{y} = Ax + w$: each row of A records an intervention, x is the map we want to recover, and w collects what the linear prediction misses. We call the design and analysis of these measurements mechanistic tomography.

Writing the problem this way gives methods with different targets a common measurement language. Coordinate patching measures finite one-component effects, attribution patching returns a local gradient map, subset interventions give aggregate forward measurements, and Hessian-vector products measure local curvature [4, 32, 38, 39]. Although their targets differ, each method requires a declared map, a measurement procedure, an intervention family, an account of the residual, and a way to check the result. Sec. 2 states this common form and derives the conditions under which its measurements are useful.

The common form also shows how an existing method can do more. If an effect map is sparse or compressible, aggregate measurements can replace exhaustive one-component probes. A few finite measurements can test whether a local gradient map needs only a simple correction at the intended intervention scale. If a structured error remains on held-out interventions, the next measurement must add interactions, measure curvature, or change the basis. Calibration dimension names the point at which a simple correction stops being enough; it is one decision inside this larger procedure. Sec. 2.1 develops this logic. It matters in real transformers because backup, suppression, redundancy, and self-repair make the effect of one component depend on the state of others [19–21].

An estimator acts as an *observer* when its output guides an intervention.¹ An observer must do more than reconstruct

¹We borrow *observer* from control theory, where it estimates unmeasured state from available inputs and outputs [41]. Here the role also includes estimators of finite intervention effects. A safety monitor is one instance of this broader observer role; its adequacy is judged relative to a specified downstream decision, not predictive accuracy alone.

a mechanism: it must predict what an intervention will actually do at the scale where it will be used—a requirement Sec. 2 develops. Control makes observer error visible because it can change the action taken, and successful target control can still move an unwanted state. Sec. 4 tests both claims while holding the model, controller, and actuator fixed where required. This use of control follows recent work that evaluates interpretability through the interventions it enables and the evidence supporting them [6, 26].

The experiments follow the same decision sequence: test how observer error affects control performance, establish the two first-order branches, then escalate to interactions, the basis, and pretrained models where complete ground truth is no longer available [17, 19, 27]. As Sec. 3 explains, mechanistic tomography starts after a candidate basis and observer family have been proposed, and it tests those choices rather than discovering the circuit or choosing the controller’s objective.

Contributions

- (1) **Structural unification.** Table 1 places coordinate patching, attribution patching, subset recovery, Hessian-vector products, and lifted recovery in one measurement language. The methods keep their own targets and numerical answers; their recoverability, error amplification, and held-out prediction error can be assessed within a common framework.
- (2) **Measurement and extension theory.** Intervention scale balances nonlinear response against amplified measurement error, while pure interactions can be invisible to first-order measurements. Lemma 1 separates nonlinear response, sampling error, and basis error; the recovery, calibration, and lifting results turn that decomposition into conditions for choosing a measurement family.
- (3) **Calibration dimension.** Proposition 1 defines the smallest declared correction family that lets a baseline map predict finite responses on held-out interventions. An r -dimensional correction needs $O(r)$ well-conditioned probes; a residual outside that family calls for new features or a new basis. Sec. 5.2 tests the dimension-one case: just eight finite probes raise held-out attribution-patching R^2 from 0.818 to 0.948 at the larger scale. Yet choosing those probes by the classical information-optimal rule gives 16.8% higher MSE than random selection in the retained-data replay.
- (4) **Control-facing validation.** In a two-HMM wind tunnel, closed-loop target error tracks observer error across ten observer variants (Spearman 0.95) under a fixed controller and actuator. A separate specificity test shows why target improvement by itself is not enough: it can coexist with increased movement of a nuisance state.

(5) Measurement savings and stopping decisions.

Twelve aggregate measurements approximately recover the HMM model’s 32-coordinate map. In the sparse synthetic testbed, 64–72 measurements recover interactions among 2,080 candidate coefficients, compared with 2,016 exhaustive pair probes. Tracr, which compiles known computations into transformers, shows that the required interaction order depends on the basis. Held-out prediction then supports adding a group interaction in GPT-2 but stopping at a calibrated additive map in Qwen-2.5-7B (Secs. 5.1–5.6).

2 Mechanistic interpretability as an inverse problem

How much a model’s output changes when we alter one or more components² depends on how we make the measurement. A finite single-component change, a local gradient, and a multi-component intervention need not return the same value because transformers are nonlinear and component effects can depend on what else changes [4, 32, 38]. We now make their common measurement structure precise for component-effect maps; Sec. 4 returns to state estimation for control.

To formalize mechanistic tomography, we first define what we observe, what we change, and over which inputs we average. Let $c \sim \mathcal{D}$ denote a model context, let h_c denote the activation state at the intervention point, and let $f_c(h)$ be the scalar output metric obtained by continuing the model from activation state h under context c . For example, f_c may be a logit difference between two candidate tokens. For each candidate component i , the direction $d_i(c)$ specifies how we intervene on that component, and $D_c = [d_1(c), \dots, d_n(c)]$ represents the set of interventions. These candidate interventions form the coordinate system in which we will estimate component effects. Choosing that coordinate system is an experimental hypothesis; it does not imply that the model’s computation is inherently organized according to those coordinates.

All of these methods begin with the same physical operation. They choose some components, push the activation along their directions, and record how much the readout moves. A method is defined by which components it changes, how far it moves them, and how it turns the measured responses into a map of component effects.

Before comparing these measurements, we need a convention for turning a choice of components into an actual activation change. Without one, a measurement that touches many components may look stronger simply because it changed

²A component is any unit that can be intervened on; examples include an attention head, MLP block, sparse auto-encoder (SAE) feature, residual-stream direction, or circuit edge.

more of them. Let z be a mask that activates an expected fraction ρ of the coordinates. We choose a normalization $s_\rho > 0$ and write $a = z/s_\rho$ for the coefficients applied to the component directions. For example, $s_\rho = \sqrt{\rho n}$ makes the rows of the design approximately unit norm. The scalar ε sets the overall size of the push. We divide the observed change by ε so that measurements taken at different scales are reported in the same units:

$$\tilde{y}(a) = \frac{1}{\varepsilon} \mathbb{E}_c [f_c(h_c + \varepsilon D_c a) - f_c(h_c)]. \quad (1)$$

Each measurement returns one observed number, $\tilde{y}(a)$. Up to an error term, that number is a weighted sum of the unknown component effects, and the weights are the entries of a . The row a says which components the measurement asked about and how strongly it asked. The unknown map x is what we want to recover. The error term records whatever the weighted sum misses. Methods differ in three ways: how they construct the row, which map they call the target, and what kind of model access produces the measurements.

Coordinate patching puts one nonzero entry in the row. It changes one component at a chosen finite scale and reads that component's finite effect directly. A subset intervention fills several entries at once. No single measurement isolates one component, but several rows together constrain the map. We then solve for the individual effects instead of reading them off one at a time.

Shrinking ε toward zero changes the target. The finite response becomes a derivative, and the target becomes the local first-order map

$$x_i = \mathbb{E}_{c \sim \mathcal{D}} [\partial_{\delta} f_c(h_c + \delta d_i(c))|_{\delta=0}], \quad (2)$$

which attribution patching estimates. This is the direct-access case of the inverse problem assembled below: the backward pass returns the coordinates of x directly, so no inversion is required. Coordinate patching at finite ε uses the same one-coordinate design but returns the finite singleton map instead. For a one-hot row e_i , $\tilde{y}(e_i) = x_i + w(e_i, \varepsilon)$, so its difference from the local map is exactly the residual we need to measure.

The measurement equation becomes useful for AtP in the forward direction. A gradient supplies x but no information about w . An intervention acts at finite scale, where w determines whether x still predicts the response. Measuring that residual requires finite measurements; Sec. 5.2 uses calibration probes for this purpose.

This forward direction is the one an observer exercises. An estimate used to explain a mechanism is checked against that mechanism. An estimate used to guide an intervention must predict the response to an action that has not yet been taken, at the scale the action will use. That second test makes

w a quantity to measure and makes ε and ρ part of the measurement design. Sec. 4 makes the control loop explicit. The formulation itself does not require a loop, but the intended use determines which residuals matter.

Two further choices stay within the same form. Adding the products $a_i a_j$ to each row adds pair coefficients to the target, so interactions become columns of a larger linear model. A Hessian-vector product along a chosen direction instead measures local curvature. It keeps the same row-times-unknown structure but replaces the effect map with the local curvature map H_0 .

Stacking m observed responses gives the standard inverse-problem form used in network tomography and compressed sensing [8, 11, 36]:

$$\tilde{y} = Ax + w. \quad (\text{MT})$$

Here $\tilde{y}$ is the vector of responses we observed, row j of A records the weights used in measurement j , and x is the hidden map we want to recover. The residual w collects whatever the weighted sum misses. For a first-order map, this includes curvature, finite-sample error, and error from the chosen basis. One feature differs from a textbook inverse problem: w is not independent noise added to a fixed system. It depends on the design because changing A changes both what we measure and what we do to the model.

The common form unifies the design questions these methods face. Each uses measurements to recover a hidden map, so each must establish whether its design contains enough information, how stable recovery is, how many measurements it costs, and what error remains on held-out interventions. The methods need not use the same basis, recover the same map, or require the same model access. A local gradient, a finite singleton effect, and a finite pair effect may have different numerical values.

Once we choose a basis, a linear inverse model gives the simplest useful prediction: a weighted sum with known weights. That form shows whether we have enough independent measurements, whether recovery is stable, whether sparsity can save measurements, and how much the procedure costs. It does not claim that the transformer itself is linear. Anything the weighted sum cannot represent appears in the residual. If that residual remains large on held-out measurements, the experiment must add interactions, change the basis, or use a richer measurement model.

Table 1 makes this unification concrete by mapping each method to its target, access path, and induced design.

Table 1: Mechanistic tomography maps each method to a target, access path, and measurement design.

Method and access	Declared target	Induced measurement
Coordinate patching; forward	finite singleton effects	one-coordinate rows
Subset recovery; forward	additive finite map	multi-coordinate rows; joint fit
AtP; backward	local first-order map x	gradient returns coordinates
Lifted recovery; forward	finite main and pair effects	columns include a_i and $a_i a_j$
HVP; white-box second order	local curvature H_0	directions give curvature projections

2.1 What the formulation lets us decide

The inverse formulation separates three decisions that arise in any measurement-based study of mechanism. First, does the proposed linear relation describe finite interventions at the scale where the estimate will be used? Second, if it does, which measurement design identifies the target with acceptable cost and error? Third, if the relation fails on held-out interventions, can a small calibration repair it, or must the measurement family be enlarged? These are the natural questions because they are the three stages of an inverse measurement procedure: model the measurements, recover the target, and diagnose the model when it fails.

A map should be tested in the operating region where it will be used. For an observer, that region is set by the intervention scale and the contexts the controller will encounter. Held-out finite interventions test whether the map is adequate for action.

An estimate can be judged by how faithfully it recovers the declared target or by how well it supports the intended action, and the two often agree. They can come apart because a downstream use weights errors unevenly. Error in directions the actuator cannot express may have little effect on the loop, while error along its effective direction matters directly. Reconstruction accuracy remains the right criterion for explanation. When an estimate will drive an intervention, reconstruction and control performance should be reported separately. Sec. 4.1 gives a case in which they diverge. This distinction also appears in network tomography [29].

The recovery tools come from classical inverse problems, sparse recovery, masked interaction recovery, and HVP correction [4, 7, 15, 39]. The new element is that each measurement is also a finite intervention on a nonlinear model, so the design changes the residual it must control.

Does one finite intervention produce a linear measurement?

The inverse formulation would be empty if we simply assumed that every intervention followed it. We instead derive the relation from the response to one finite intervention. The lemma below shows that a mask a produces a weighted sum

of the local component effects, $a^\top x$, plus three identifiable sources of error.

LEMMA 1 (MECHANISTIC MEASUREMENT REDUCTION). *Let $\varepsilon > 0$ and let f_c be twice differentiable on the perturbation ball containing the intervention paths. Assume its Hessian satisfies $\|\nabla^2 f_c\|_{op} \leq L$. Define the population normalized response*

$$y_\infty(a, \varepsilon) = \frac{1}{\varepsilon} \mathbb{E}_c [f_c(h_c + \varepsilon D_c a) - f_c(h_c)]. \quad (3)$$

Suppose the response estimated from a batch of B paired contexts can be written as

$$\tilde{y}_B(a, \varepsilon) = y_\infty(a, \varepsilon) + \frac{e_B(a)}{\varepsilon} + \zeta_B(a, \varepsilon), \quad (4)$$

where $|e_B(a)| \leq \tau_B(a)$ bounds uncanceled error in the response numerator and $|\zeta_B(a, \varepsilon)| \leq \nu_B(a)$ bounds the remaining normalized sampling error. Then

$$\tilde{y}_B(a, \varepsilon) = a^\top x + w(a, \varepsilon), \quad (5)$$

where x is the local first-order map in Equation (2) and

$$|w(a, \varepsilon)| \leq \frac{L}{2} \varepsilon \mathbb{E}_c \|D_c a\|_2^2 + \frac{\tau_B(a)}{\varepsilon} + \nu_B(a). \quad (6)$$

Stacking measurements from several masks gives $\tilde{y} = Ax + w$. The proof is in Appendix A.

Suppose a mask changes only heads 2 and 5, with coefficients a_2 and a_5 . The linear part of the measured response is

$$a^\top x = a_2 x_2 + a_5 x_5.$$

The intervention returns one weighted sum of the effects without revealing either separately. Several masks produce several such equations, and solving those equations recovers the individual effects. This is the same logic used in network tomography and compressed sensing.

The residual w records three sources of difference between the measured transformer response and that weighted sum. The curvature term grows with ε because a larger intervention moves farther from the local tangent and encounters more of the model's nonlinear response. The term τ_B/ε grows when ε becomes too small because any error already present in the response numerator is magnified when we divide by the intervention size. The remaining term ν_B records normalized sampling error that remains after pairing or batching.

The lemma therefore rules out both extremes: a very large intervention lets curvature dominate, and a very small one magnifies whatever numerator error remains. Pairing the clean and intervened evaluations helps because it can cancel shared variation before division by ε .

The mask affects more than the row of A . It also changes the displacement $\|D_c a\|$, the amount of curvature encountered, and potentially the sampling error. The design matrix and the residual are therefore coupled. This is the main

difference from a textbook linear inverse problem with independent noise added after the measurements have been chosen.

The lemma establishes the local measurement model and makes the design dependence of its residual explicit. Recovery then depends on the rank, coverage, or sparse-recovery properties of A . The next results connect these properties to the choice of scale and density and to the decision to calibrate or enlarge the measurement family.

Can a small correction repair the finite response? For a non-differentiable endpoint—for example, a behavioral rating, tool outcome, or black-box score—we cannot use the Taylor bound directly. In that setting the finite response is the quantity of interest, and held-out interventions provide the relevant test. The remaining question is whether the difference from the gradient map is a simple scale correction or evidence that the first-order family is missing structure. Calibration dimension formalizes that distinction.

A useful first check is whether all finite effects are approximately a rescaled version of the local map. If so, a few finite probes may be enough to estimate the correction. If not, collecting more probes for the same family will not solve the problem.

PROPOSITION 1 (CALIBRATION DIMENSION). *Let $F_\epsilon(a)$ denote the systematic finite-intervention response on a design family $\mathcal{A}$, and let Q be a declared held-out distribution on that family. Fix a declared feature map $\phi(a) \in \mathbb{R}^q$, baseline $x_0 \in \mathbb{R}^q$, and correction dictionary $B \in \mathbb{R}^{q \times r}$. If*

$$F_\epsilon(a) = \phi(a)^\top (x_0 + B\theta_\epsilon) \quad \text{for all } a \in \mathcal{A} \quad (7)$$

and the sampled correction design ΦB has full column rank, then θ_ϵ is identifiable from $O(r)$ noiseless calibration measurements and stably estimable according to the conditioning of ΦB , where row j of Φ is $\phi(a_j)^\top$. In all cases, the residual

$$\inf_{\theta} \|F_\epsilon(\cdot) - \phi(\cdot)^\top (x_0 + B\theta)\|_{L_2(Q)} \quad (8)$$

lower-bounds the error of every observer using that family.

For a declared nested family B_r and a tolerance δ at or above the estimated noise floor, the calibration dimension is the smallest r whose held-out error is at most δ . Dimension zero means that the baseline map transfers without correction. Dimension one allows one scalar gain. Larger values allow several stated corrections, such as separate gains for component groups. If the residual lies outside the entire first-order family, no collection of gains can fix it; the observer needs new features, such as component pairs. Sec. 5.2 tests the one-gain case; Sec. 5.3 supplies a case in which pair features are necessary.

This definition gives a practical stopping rule. Start with a gradient map when a backward pass is available; otherwise

start with an additive map estimated from forward interventions. Fit the correction on one set of masks and evaluate it on another. The held-out split matters because curvature and correlated masks can make an inadequate family fit its training measurements. In the ideal noiseless case, an r -dimensional correction needs only $O(r)$ well-conditioned probes instead of re-estimating n main effects or $O(n^2)$ pairs. Move to a richer family only when the held-out residual remains above the estimated noise floor and the richer family improves it reproducibly.

How should scale and mask density be chosen? After choosing an observer family, we still need to choose the interventions used to measure it. Two design variables matter immediately. The scale ϵ controls how far each probe moves the model, while the density ρ controls how many coordinates each mask changes. Small probes reduce nonlinear error but magnify uncanceled response noise. Dense masks may cover sparse features more efficiently, but they can also move the activation farther or make the inverse problem poorly conditioned.

COROLLARY 1 (NOISY FIRST-ORDER SCALE-DENSITY BOUND). *For a mask family $\mathcal{A}_\rho$ under a fixed actuator normalization, define*

$$\kappa_\rho^2 = \sup_{a \in \mathcal{A}_\rho} \mathbb{E}_c \|D_c a\|_2^2, \quad (9)$$

and let $\tau_B(\rho)$ and $v_B(\rho)$ uniformly bound the two sampling terms in Lemma 1. For the declared target class, assume the recovery method obeys $\|\hat{x} - x\|_2 \leq C_{\text{rec}}(A_\rho) \|w\|_\infty$ when $\tilde{y} = A_\rho x + w$. Thus C_{rec} amplifies the maximum normalized measurement residual into reconstruction error. A conditional first-order recovery-error bound is

$$\mathcal{J}_1(\epsilon, \rho) = C_{\text{rec}}(A_\rho) \left[\frac{L}{2} \epsilon \kappa_\rho^2 + \frac{\tau_B(\rho)}{\epsilon} + v_B(\rho) \right], \quad (10)$$

$$(\epsilon^*, \rho^*) \in \arg \min_{\epsilon, \rho} \mathcal{J}_1(\epsilon, \rho).$$

For fixed ρ , if $\tau_B(\rho) > 0$, $L\kappa_\rho^2 > 0$, and C_{rec}, v_B are locally independent of ϵ , the first two terms have the interior minimizing scale

$$\epsilon^*(\rho) = \sqrt{\frac{2\tau_B(\rho)}{L\kappa_\rho^2}}. \quad (11)$$

Density affects both model-induced error through κ_ρ and identifiability through coverage and conditioning.

Corollary 1 gives a square-root rule for the interior scale that minimizes the stated error bound, balancing curvature against amplified numerator error [31]. Secs. 5.1–5.3 vary scale, budget, and noise separately to examine these trade-offs. Using the bound to select a joint scale-density design additionally requires estimates of L , τ_B , and design conditioning.

Appendix D gives the scale–density proof and explains how second-order targets change the normalization.

How the mask family changes the design. The mask family affects both activation displacement and recovery conditioning. Under fixed per-coordinate amplitude, a denser mask usually moves the activation farther; under unit-row normalization, the perturbation is spread over more coordinates. The distinction matters because τ_B describes numerator error amplified by division by ε , whereas ν_B describes the paired sampling error left after normalization. For pair or Hessian targets, some quadratic response becomes signal, while mask density still determines whether the pair features receive enough independent coverage.

A bias-aware measurement criterion. For full-rank calibration and small effect maps, we can also track how each probe’s error propagates into the estimate. Appendix E writes mean squared error as sampling variance plus squared propagated bias and, under the lemma’s Hessian bound, bounds that bias for each candidate intervention. The resulting criterion compares designs under a fixed budget, counting acquisition of curvature and noise estimates in the cost. Pairing an additive push with its negation cancels shared quadratic curvature; a bounded third derivative gives a better worst-case rate. This full-rank criterion complements the sparse-recovery guarantees below.

Sparse-recovery consequences. If the component-effect map is sparse or compressible, aggregate masks can recover it with fewer forward evaluations than patching every coordinate. Once the measurement relation is valid, basis pursuit and orthogonal matching pursuit (OMP) use the standard sparse-recovery guarantees, provided that the mask design is sufficiently well conditioned [5, 8, 34]. Coordinate-only forward patching has a simple worst-case limitation: with fewer than n measurements, at least one coordinate remains unmeasured, and the entire effect may be hidden there. This bound is worst-case and applies to forward-only recovery; a white-box gradient returns all first-order coordinates in one backward pass.

Why first-order maps miss interactions. Sparse recovery cannot recover a term that the measurement model does not contain. This matters for real transformers. Softmax attention, layer normalization, nonlinear MLPs, and downstream heads all allow the effect of one component to depend on the state of another. Backup heads may compensate for an ablated primary head; self-repair can reroute a computation; and copy-suppression heads act conditionally on what earlier heads have written [19–21]. An additive model cannot express these dependencies. More first-order measurements

only estimate the wrong family more accurately. Pair features are the smallest extension that can represent a two-component interaction.

PROPOSITION 2 (FIRST-ORDER BLINDNESS TO PURE INTERACTIONS). *Let $H_{0,ij} = \partial^2 f / \partial d_i \partial d_j$ at the baseline. If $x_i = x_j = 0$ but $H_{0,ij} \neq 0$, gradients and singleton first-order measurements do not identify the cross term. For the following finite-scale expansion, additionally assume that f has a uniformly bounded third derivative on the intervention paths. At scale $\varepsilon > 0$, the normalized finite response has*

$$\bar{y}(a) = a^\top x + \sum_{i < j} a_i a_j \Gamma_{\varepsilon,ij} + w_\varepsilon(a), \quad (12)$$

where $\Gamma_{\varepsilon,ij} = \varepsilon H_{0,ij} + O(\varepsilon^2)$. Lifted features $a_i a_j$ make these finite pair coefficients visible.

The lifted model treats each product $a_i a_j$ as an additional measurement feature. It can make a pair visible, but it also creates many more columns: n main effects become $n + \binom{n}{2}$ main-and-pair effects. A design that is well conditioned for the original n columns may be badly conditioned after this expansion.

PROPOSITION 3 (CONDITIONAL LIFTED-RECOVERY REDUCTION). *Let $\theta_\varepsilon = (x_\varepsilon, \text{vec } \Gamma_\varepsilon) \in \mathbb{R}^N$ be sparse in a lifted finite main-plus-pair basis, and let $\Phi(A)$ be the corresponding lifted design with rows $[a, \{a_i a_j\}_{i < j}]$. If the lifted misspecification is bounded, sparse recovery gives the same form of stable error bound as classical sparse recovery, conditional on restricted conditioning of $\Phi(A)$ on the relevant support. A first-order restricted-isometry property (RIP) does not imply lifted RIP. Appendix B gives the reduction to the standard sparse-recovery theorem.*

With ideal dense independent Rademacher masks, whose entries are independent random $+1$ or -1 values, the main-effect columns a_i and pair columns $a_i a_j$ are orthogonal in the population. Real designs are finite and often sparse or constrained. Pair columns can then duplicate or nearly duplicate one another. We must inspect the conditioning of the lifted design and test the recovered map on masks that were not used for fitting.

Appendix C gives a three-component half-fraction design with orthogonal main effects but aliased pair effects.

Held-out prediction is part of identification. On the training masks, a true pair and a false pair can produce the same column and appear equally plausible. A separate or explicitly de-aliased mask design tests whether the recovered interaction transfers.

How should missing interactions be measured? White-box access solves the first-order part cheaply, but one backward pass still does not return the full pair map. We need measurements of curvature as well. A Hessian-vector product

(HVP) measures how the gradient changes along a chosen direction without constructing the full Hessian, so designed HVP directions can serve as aggregate measurements of local interactions.

COROLLARY 2 (WHITE-BOX SECOND-ORDER TOMOGRAPHY). *A backward pass gives ∇f , while designed HVP queries give linear measurements of the local curvature map H_0 . If the response is quadratic on the intervention path, the known relation $\Gamma_\epsilon = \epsilon H_0$ converts local curvature to the normalized finite pair map. Otherwise HVPs identify only a local quantity; finite-scale prediction needs path integration or finite calibration. Lifted forward measurements estimate the finite-scale map directly. Integrated Hessians and HVP correction provide precedents [14, 39].*

The access regime again determines the efficient route. With gradients and HVPs, we can estimate local main and pair effects without evaluating every finite mask. With only forward evaluations or with a nondifferentiable endpoint, lifted subset interventions measure the finite pair response directly. The two routes target the same quantity only when local curvature transfers to the finite intervention scale; otherwise we must integrate along the path or calibrate against finite responses.

Appendix D gives the proofs of the calibration, scale-density, interaction-blindness, and HVP results.

3 Choices that precede recovery

The researcher chooses the basis that defines the coordinates of A and x , states what x is meant to estimate, and specifies the context distribution over which the measurements should transfer. The inverse formulation tests these choices once they have been made. The experiments below place each choice under control before considering settings with fewer external references.

The basis. Layers, attention heads, MLPs, residual subspaces, SAE features, and token positions are all plausible coordinates. They make different claims about the units of the mechanism. A mechanism that is sparse in one basis can look distributed in another. A poorly chosen coordinate can also track the output without playing the proposed causal role. Basis selection is therefore closer to dictionary learning than to regression with known features [3]. Superposition and the non-uniqueness of SAE dictionaries mean the choice changes what is measured [9, 16]. Secs. 5.3 and 5.4 put this choice under control: the planted testbed fixes the basis by construction, while Tracr lets us vary the basis without losing knowledge of the computation.

The target. A direction can change behavior without representing the variable we think it represents. For example, an

entropy-aligned direction may predict uncertainty and provide a useful steering axis even if it is not the model’s belief state or a causal bottleneck [2]. Successful control does not identify the mechanism by itself. We need an independent reference, such as an analytic posterior, a planted or compiled circuit, or a causal intervention test [1, 12, 13]. Sec. 4.1 supplies such a reference and asks how observer error affects control performance.

The context distribution. Equation (2) averages effects over a declared context distribution. If the local response changes direction across prompts, one average can mix distinct mechanisms and describe none of them well. A component-effect map is tied to the contexts used to construct it. The held-out evaluation must preserve the variation expected when the observer is used. Sec. 5.5 returns to this issue in a pretrained model, where the ranking of interaction terms is stated for a specific intervention and prompt distribution.

The experiments place these three objects under control and then relax that control one step at a time. The HMM supplies an exact reference for the target quantity. The planted testbed fixes both the basis and the nonzero support. Tracr keeps the computation known while the basis changes. IOI retains documented component groups and a declared intervention distribution, but no complete mechanism ground truth. Each step gives up one kind of external reference only after the preceding measurement question has been checked.

4 How does estimation error affect control performance?

Before testing the operational extensions, we run two control checks. First, we hold the transformer, actuator, and controller fixed and vary only the observer, which isolates how observer error affects control performance. Second, a target–nuisance estimate drives a correspondingly mixed actuation direction, which asks whether target improvement can hide nuisance movement. Here an *observer* is any estimate that guides an intervention.

A control loop connects the system being changed, the observer, the controller, and the actuator. The transformer has an internal activation state x_t ; in classical control terminology, it is the *plant*.³ The quantity we want to regulate is $z_t = \phi(x_t)$, but the controller does not observe it directly. It receives measurements y_t , uses an observer O to form the estimate $\hat{z}_t$, and then applies a control rule K :

$$z_t = \phi(x_t), \quad y_t = h(x_t, c_t) + \epsilon_t, \quad (13)$$

$$\hat{z}_t = O(y_{0:t}, c_{0:t}), \quad u_t = K(\hat{z}_t, r_t). \quad (14)$$

³More precisely, the controlled system includes the transformer state at the edit location and the remaining computation from that state to the readout. We use *transformer* in the text and reserve *plant* for this classical control-theory meaning.

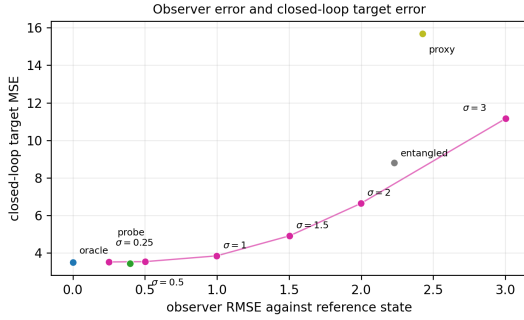


Figure 1: With the controller and actuator fixed, closed-loop target error rises with observer error across ten observer variants. The analytic oracle and the linear probe have the lowest estimation and control error; coarse, noisy, proxy, and entangled estimates have more of both.

Here c_t denotes the context, ϵ_t measurement noise, r_t the desired reference, and u_t the intervention. In an activation-steering example, x_t is the residual-stream state, O is a probe or another state estimate, K converts the estimated error into an edit size, and the actuator is the direction along which the activation is changed. The intervention could instead be a logit bias, retrieval insertion, prompt rewrite, or decoding change. Even a well-behaved controller K can fail if the observer O estimates the wrong quantity. We validate the estimate in four ways: agreement with an oracle when one exists, sensitivity under a fixed controller, movement of nuisance variables, and prediction on held-out interventions in a real model.

4.1 Estimation error changes control performance

We need a setting in which the quantity an observer should recover is known exactly. A hidden Markov model (HMM) supplies that reference: after every token, we can compute the exact filtered posterior over its changing hidden state. This lets us measure observer error before placing the estimate in a feedback loop. We start with an HMM because a natural-language task does not provide the model’s true internal belief as a reference.

Building on analytic-posterior HMM wind tunnels [1], we train a small transformer on tokens generated by two independent HMMs. One hidden process defines the target belief z_1 ; the other defines a nuisance belief z_2 . We use their exact posterior log-odds as the two reference states. The model matches the Bayes cross-entropy floor, and a linear probe recovers z_1 with $R^2 \approx 0.99$.

The first check asks whether observer quality matters once the estimate enters a feedback loop. We create ten observers

with different amounts and kinds of error, including a last-observation proxy that uses only the current emission instead of the filtered history. We then hold the transformer, actuator, controller, and target fixed. Across these observers, closed-loop target error rises with observer RMSE (Spearman $r_s = 0.95$). This fixed-controller comparison connects estimation accuracy to downstream performance in a setting with an exact reference state.

The second check asks whether an observer can improve the target while moving an unrelated state. We construct an entangled estimate, $z_1 + 0.5z_2$, and use the corresponding mixed actuation direction $d \propto d_1 + 0.5d_2$. This condition reduces target MSE from 25.02 without control to 4.81, while the oracle target-only condition gives 3.56. Yet nuisance movement rises from 0.037 to 0.079. This joint observer-actuator stress test shows that specificity matters even when the target improves, since the mixed intervention also moves the nuisance state. Reporting both quantities reveals an effect that target MSE alone would conceal.

The most faithful estimate need not control best. The two criteria mostly agree here, and their divergence explains why both are useful. The linear probe sometimes produces slightly lower closed-loop error than the analytic posterior oracle, even though the oracle is the more faithful estimate of the external HMM posterior. The controller edits the state represented by the transformer, and the probe can align more closely with that editable state than the external posterior does. We report both measures because they distinguish reconstruction from control utility: the analytic posterior tests representational faithfulness, while closed-loop performance tests how well an observer matches the state used by this model-actuator pair.

These results motivate measuring an adequate observer as cheaply as the access regime allows. We keep the model and task fixed and begin with first-order measurements under forward-only and white-box access.

5 From exact ground truth to pretrained LLMs

5.1 Aggregate measurements replace exhaustive probes when the map is compressible

Without gradient access, coordinate patching requires a separate forward intervention for every coordinate. Mechanistic tomography suggests changing several coordinates together and treating the response as one aggregate measurement. Each such intervention supplies one row of A . If only a few coordinates have large effects, several aggregate rows may recover the map with fewer interventions than probing every coordinate separately.

We test this possibility in the same HMM model used for the control experiment. We define 32 coordinates by crossing transformer layer with token position. This dimension is small enough that we can also patch all 32 coordinates individually. Their finite effects provide a reference for evaluating aggregate recovery.

The measured response is the change in the model’s implied belief log-odds at the final token, averaged over a batch. We invert its predicted next-observation probability using the HMM parameters, clip the implied belief probability to $[10^{-5}, 1 - 10^{-5}]$, and then take log-odds. This output-derived readout is distinct from the analytic posterior reference. Because paths crossing the clipping boundaries fall outside Lemma 1’s smoothness assumptions, these finite-effect experiments test the measurement equation empirically.

The reference map is not exactly sparse, but it is strongly compressible, with its four largest coordinates containing 98.4% of its squared ℓ_2 norm. We use orthogonal matching pursuit (OMP), a sparse-recovery method that selects a small set of coordinates and fits their effects from the aggregate responses. A separate validation split chooses how many coordinates OMP should retain. We also fit ridge regression as a dense baseline.

After 12 aggregate measurements, validation selects a four-coordinate OMP model. It gives Pearson $r = 0.989$ against the coordinate-patching reference and $R^2 = 0.935$ on held-out aggregate masks. Ridge regression requires 32 measurements, the same count as exhaustive coordinate patching, to obtain $r = 0.962$ and held-out $R^2 = 0.933$.

Aggregate measurements save interventions here because the finite-effect map is compressible and approximately additive. These two properties determine when the saving is available; dense maps or strong interactions call for a different design. Bair et al. demonstrate the same sparse-recovery principle at pretrained-model scale using attention-head knockouts [4]. This experiment establishes the forward-only baseline before we turn to gradient access.

5.2 A few finite probes calibrate attribution patching at intervention scale

With gradient access, attribution patching returns the complete local first-order map in one backward pass. This is cheaper than recovering the same map from many forward measurements. The limitation is that a gradient describes an infinitesimal change, while an actual intervention has finite size. Nonlinearities in either the model or its readout can make the finite response differ from the gradient prediction.

We test whether a single scale factor can correct the mismatch before estimating a richer map. Using the same HMM model and component directions as in Sec. 5.1, we compare scales $\varepsilon \in \{5, 8\}$. Raw attribution patching predicts the

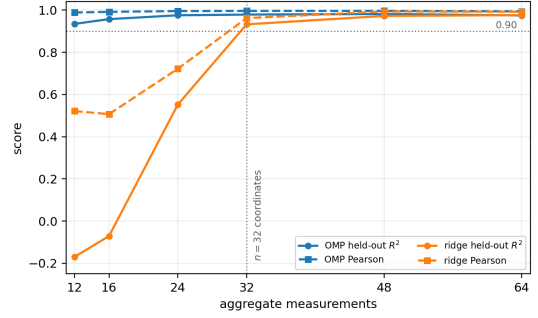


Figure 2: Aggregate recovery of the 32-coordinate finite-effect map. OMP scores Pearson $r = 0.989$ and held-out $R^2 = 0.935$ with 12 aggregate measurements. Ridge gives comparable held-out prediction with 32 measurements, the exhaustive coordinate count.

Table 2: Finite-scale calibration on one trained model whose intervention directions are estimated once. We cross three independently generated 256-sequence evaluation batches with three mask-and-split designs. Entries are means [min, max] across the nine cells.

Scale	raw AtP R^2	calibrated AtP R^2	OMP R^2	fitted gain $\bar{g}$
$\varepsilon = 5$	0.943 [0.920, 0.967]	0.963 [0.933, 0.983]	0.965 [0.952, 0.976]	0.874 [0.799, 0.915]
$\varepsilon = 8$	0.818 [0.751, 0.903]	0.960 [0.925, 0.978]	0.969 [0.960, 0.977]	0.742 [0.677, 0.780]

smaller interventions well on average, while it systematically overpredicts responses at the larger scale. With the readout defined in Sec. 5.1, the gain corrects the combined model-and-readout response. We fit one scalar gain from finite calibration probes and rescale the gradient map, preserving its relative sensitivities without re-estimating every coordinate.

To check that the gain transfers across sequences and measurement designs, we cross three independently generated evaluation batches with three independently generated mask-and-split designs, giving nine evaluation cells. Each cell contains 128 masks: 72 fit the scalar gain and the sparse OMP comparison, 24 select the OMP model, and 32 are reserved for final evaluation. We fit a separate gain at each intervention scale because the correction describes the response at that scale.

At $\varepsilon = 8$, calibration raises mean held-out R^2 from 0.818 to 0.960, close to OMP’s 0.969. A gain fitted on the other two evaluation batches gives a leave-one-batch-out mean of 0.959 across the nine batch-and-design cells. The correction transfers across sequence batches. At $\varepsilon = 5$, where raw AtP is already strong, calibration raises mean held-out R^2 from 0.943 to 0.963, compared with 0.965 for OMP.

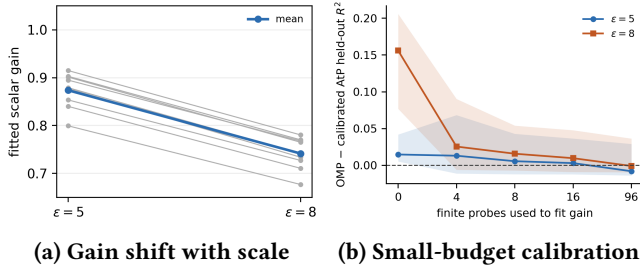


Figure 3: Finite-scale calibration across the fixed-direction 3×3 design. Left: fitted gain at each scale for the nine evaluation-batch and mask-design cells. Right: the median held-out R^2 gap between OMP and calibrated AtP as the calibration budget grows; bands show the 10th–90th percentiles over cells and subset draws.

We next ask how many finite probes the correction needs. From the 96 masks not reserved for final evaluation, we repeatedly draw calibration sets of 4, 8, and 16 probes; the 96-probe condition uses the full pool. Eight probes already give mean held-out $R^2 = 0.951$ at $\epsilon = 5$ and 0.948 at $\epsilon = 8$. Sixteen probes raise these values to 0.957 and 0.955 . Calibration is the first extension to try when gradients are available, since a handful of finite measurements can correct most of the scale error without recovering the component map again.

The scalar gain tests the calibration-dimension-one hypothesis. It corrects most of the scale error, while the residual gap to OMP in some cells, even after 96 probes, identifies where one gain leaves useful predictive structure unexplained. Held-out error thus supplies the stopping rule: retain the corrected map when it predicts the intended interventions well enough, and use a richer observer when the residual warrants it.

Appendix E.5 compares three fixed ways to select calibration probes from the retained measurements. Its bias-variance decomposition explains why selecting the largest predicted responses can increase held-out error despite the rule’s variance advantage under a classical linear error model.

5.3 Interaction measurements recover effects that first-order maps miss

Scalar calibration can change the size of a first-order effect, but it cannot create an effect that appears only when two components change together. If the response contains a term such as $a_i a_j$, no gain applied to an additive map can represent it. This matters for transformers because backup, suppression, redundancy, and self-repair make the effect of one component depend on whether another is active. Before studying these effects in a pretrained model, we use a planted system in which the correct terms are known. This

lets us distinguish three failures that can otherwise look alike: choosing a family that cannot represent the response, collecting too few measurements, and selecting a correlated but noncausal feature.

The testbed contains 64 candidate components. We plant four nonzero terms: two ordinary main effects, one pure interaction with no first-order signal, and one pair that models redundancy or self-repair. We can also add an off-path confound. Designed-mask methods such as SPEX and ProxySPEX use the same general principle to recover sparse feature and attention-head interactions [7, 15].

To represent pair effects, we expand each measurement row. In addition to the original coefficients a_i , the row now contains every product $a_i a_j$. The unknown map has one coefficient for each component and one for each unordered pair:

$$N = n + \binom{n}{2} = 64 + 2016 = 2080. \quad (15)$$

We call this the *lifted* basis because pair products become ordinary columns in a larger linear model. Only four of the 2,080 possible terms are nonzero. Measuring every pair separately would require 2,016 pair probes. Aggregate masks may recover the same sparse map with far fewer measurements.

Can the family represent the response? We begin with a generous budget of 256 low-noise aggregate measurements. This removes measurement scarcity as a likely explanation for failure. At $\epsilon = 8$, lifted OMP recovers the planted pair structure and gives nearly perfect prediction on held-out masks. Raw AtP, scalar- and multi-gain AtP, finite singleton patching, and first-order OMP all omit pair features. Each of these additive methods gives held-out $R^2 < 0.5$.

The additive feature family lacks the cross terms required to represent this response. Fitting its coefficients more accurately or adding calibration measurements cannot repair that omission.

How many forward measurements are needed? Once we know that the lifted family can represent the response, we reduce the measurement budget. We require both held-out $R^2 \geq 0.95$ for finite-response prediction and pair recall ≥ 0.99 for the planted support. Lifted OMP first meets both conditions with 64 aggregate measurements at $\epsilon = 5$ and 72 at $\epsilon = 8$. Exhaustively measuring every pair would require 2,016 probes.

The heuristic count is $4 \log(N/4) \approx 25$ measurements. It omits constants and the effects of mask normalization, finite-sample variation, and lifted conditioning. We report the budget at which prediction and support recovery both meet their thresholds, accounting for variability across low-budget trials.

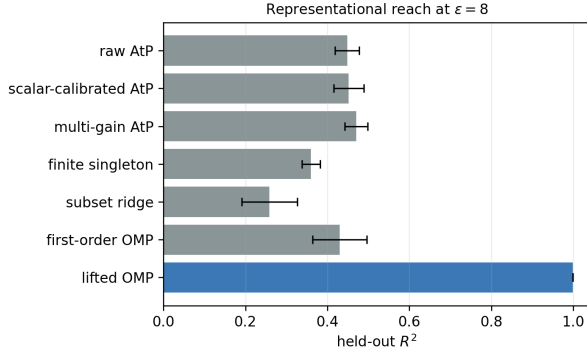


Figure 4: Interaction-aware recovery at $\epsilon = 8$ with a generous fixed budget. Lifted OMP recovers the planted pair-bearing response, while methods without pair features remain below 0.5 held-out R^2 .

White-box alternative: AtP plus HVPs. Lifted forward measurements are useful when the endpoint is available only through model evaluations, as with sampling, decoding, retrieval, tool use, or a behavioral score. With white-box access, we can measure the local pair structure more directly. A gradient asks how the output initially changes along each component direction. An HVP asks how those gradient effects change when we move in a chosen direction. Attribution patching supplies the main-effect map in one backward pass, while designed HVP queries provide aggregate measurements of pair curvature.

The planted response is exactly quadratic. For this system, multiplying the local HVP pair coefficient by ϵ gives the normalized finite-scale pair coefficient. This known conversion provides a reference for the local-to-finite step. In a transformer, path integration or finite calibration establishes whether local curvature predicts the response at the intended scale.

The combined AtP–HVP observer meets held-out $R^2 \geq 0.95$ and pair recall ≥ 0.99 with 12 HVP queries at $\epsilon = 5$ and 24 at $\epsilon = 8$. These testbed-specific budgets count HVP queries in addition to the AtP backward pass that supplies the main effects.

The pair-only HVP ablation finds the correct interacting pairs but gives $R^2 = 0.36$ at $\epsilon = 5$ and 0.60 at $\epsilon = 8$. Recovering the pair support is not enough to predict the full response. The observer must combine the AtP main effects with the HVP pair effects. Replacing the gradient main effects with finite singleton effects also performs worse. In this testbed, singleton interventions saturate and underpredict larger aggregate changes, while the gradient remains the better main-effect approximation once the missing pairs are included.

Support recovery and response prediction are different. We next use 96 forward measurements, above the clean recovery threshold, and add zero-mean Gaussian noise to each normalized response. This separates two possible goals: locating the interacting pairs and estimating their coefficients accurately enough to predict a new intervention.

At $\epsilon = 5$, both prediction and pair recall remain high through noise standard deviation 0.01, and pair recall remains high through 0.05 after prediction degrades. At $\epsilon = 8$, both remain high through 0.05, and pair recall remains high through 0.2 after prediction degrades. The interacting pair can remain identifiable after coefficient error has made the observer unsuitable for finite-response prediction.

Why held-out masks matter. Finally, we give one off-path coordinate the same pattern as a causal coordinate on the training masks. Either feature can then explain the observed responses, even though only one belongs to the planted mechanism: the two features are aliased by the measurement design.

Sparse support selection followed by independent held-out masks suppresses the false positive. The test shows how held-out interventions contribute to identification by separating features that agree on the training masks. A corresponding pair-level stress test would apply the same principle to aliased columns in the lifted design.

5.4 The basis determines the required measurement family

The planted experiment in Sec. 5.3 gives us the correct coordinates in advance, whereas real interpretability work must identify them. The same computation can look additive in a basis that contains a completed detector and interactional in a basis that contains only the detector’s inputs. We need a system in which we can change the basis while keeping the underlying computation known.

A simple conjunction illustrates the issue. Suppose the model must report whether the previous token is A and the current token is B. If the basis contains one completed “AB detected” coordinate, an additive readout can use that coordinate directly. If the basis contains only separate coordinates for “previous token is A” and “current token is B,” neither coordinate is sufficient by itself, and the readout needs their product. The computation is the same; only the coordinates used to describe it differ.

Tracr provides this intermediate setting [17]. It compiles programs written in the restricted-access sequence processing (RASP) language into executable transformers. The compiler also assigns names to residual-stream coordinates associated with program variables. This lets us inspect the same known computation through different sets of coordinates.

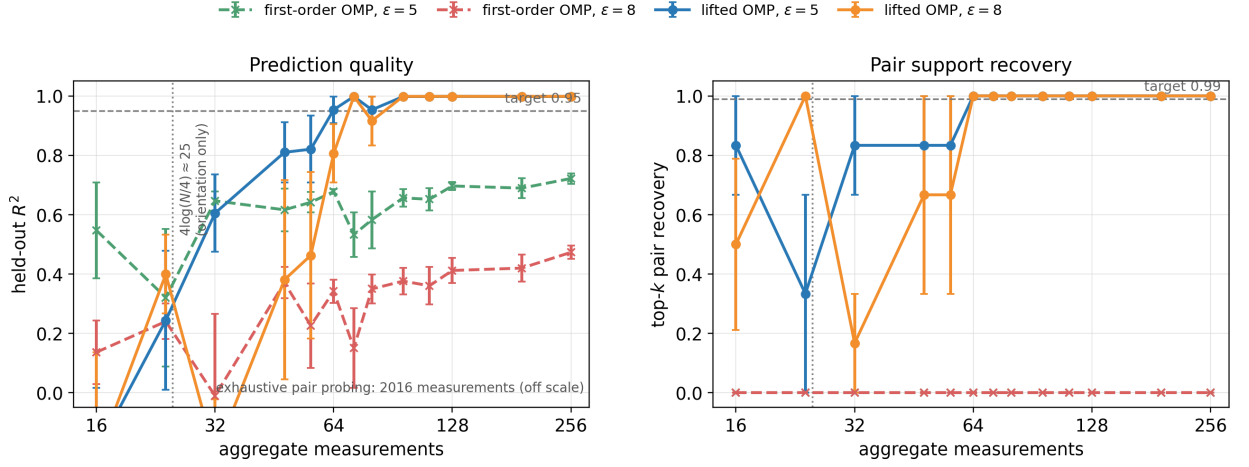


Figure 5: Measurement budget for lifted recovery. With four planted nonzero terms among 2,080 main-and-pair features, lifted OMP reaches the prediction and pair-recall thresholds with 64–72 aggregate measurements, compared with 2,016 exhaustive pair probes. First-order OMP has no pair features.

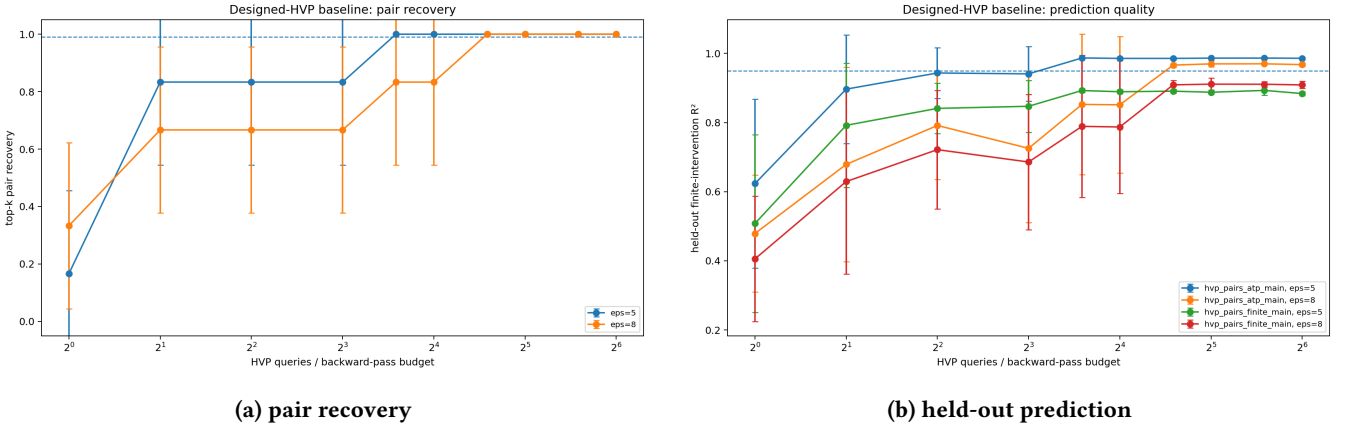


Figure 6: White-box interaction recovery. Designed HVPs recover the local pair map, and the quadratic testbed supplies the known conversion to finite-scale pair coefficients. Accurate prediction also requires the AtP main-effect map.

We compile three programs over the symbols A, B, and C. The first reports a property of the current token and serves as an additive control. The other two detect the sequence pattern “A followed by B.” One uses a numeric representation and the other a boolean representation. Both compute the same conjunction but place it in different compiler-generated coordinates.

At sequence length 6, we enumerate all 243 possible input sequences. After excluding the position with no predecessor, this gives 1,215 position-level examples. We compare two bases. The *native* basis contains the full set of compiler variables, including the completed AB detector. The *restricted* basis contains only variables available before the conjunction

has been formed, such as the current-token and previous-token labels.

Does the apparent interaction order change with the basis?

For the current-token control, an additive model is sufficient in either basis. Both the full native basis and the token-only basis give held-out position-level $R^2 = 1.0$.

The AB detectors are also additive in the native basis. This does not mean that the conjunction has disappeared. The compiler has already created an explicit coordinate that records whether “A followed by B” is true. Once that completed detector is available as one coordinate, an additive readout can use it directly.

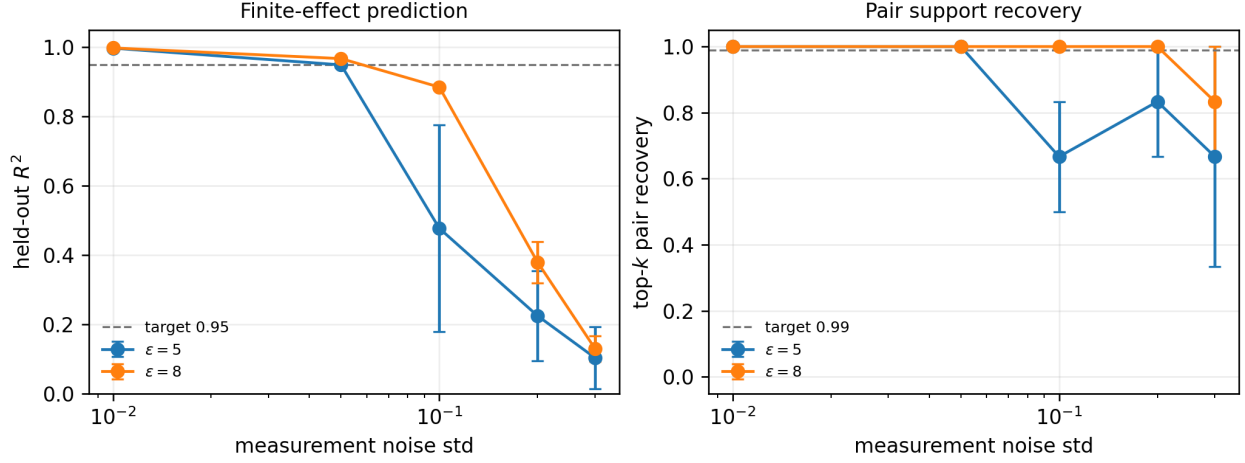


Figure 7: Noise robustness with 96 aggregate measurements. Pair support can remain recoverable after held-out finite-response prediction degrades, separating support identification from coefficient accuracy.

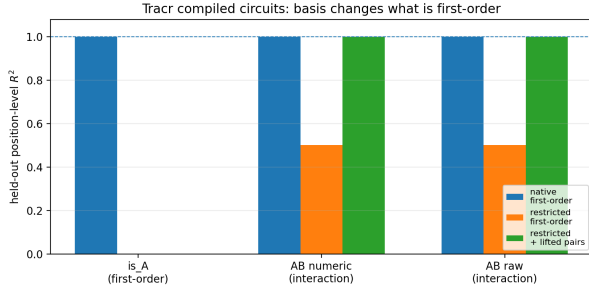


Figure 8: Basis comparison in Tracr circuits. The current-token control is additive in both bases. The AB detectors are additive in the native compiler basis because it contains an explicit detector coordinate. In the restricted pre-conjunction basis, additive features explain roughly 50% of the variance, while lifted pair features recover the intended conjunction and restore held-out $R^2 = 1.0$.

The result changes in the restricted basis. This basis contains the current-token and previous-token information but no completed AB detector. An additive model explains about 50% of the detector variance. Adding pair products restores held-out $R^2 = 1.0$ and selects the intended term:

$$\text{previous-token-A} \times \text{current-token-B.}$$

The AB computation is first-order in the native basis and interactional in the restricted pre-conjunction basis.

Does the readout use the reconstructed detector? Predicting a compiler label does not show that the model’s output depends on it. To test that dependence, we intervene in the final residual stream, immediately before the readout.

The numeric detector group contains the detector and its AB/OTHER coordinates. The boolean group contains the True/False detector and map coordinates.

We first ablate the detector group. We then reconstruct it from the restricted pre-conjunction basis using either additive features or lifted pair features and write that reconstruction back into the residual stream. This tests whether the recovered detector supplies information that the final readout actually uses.

Editing the final residual stream isolates the connection between the detector subspace and the readout. Reconstruction at an earlier layer would additionally have to survive the intervening transformer computation, so the writeback result applies to the final-residual interface tested here.

Ablating the detector group drops held-out readout R^2 from 1.00 to about -0.05 for the numeric program and -0.51 for the boolean program. The final output therefore depends on these detector coordinates. Writing back the additive reconstruction restores readout R^2 to only about 0.55 for both programs. Writing back the lifted reconstruction restores the original $R^2 = 1.00$.

The same pattern appears when we score reconstruction of the detector group itself. Additive features give detector-group reconstruction R^2 of about 0.68 for the numeric group and 0.93 for the boolean group. Lifted features reconstruct both groups perfectly. The apparently high additive score for the boolean group is misleading: it mostly captures the common negative case and misses the rare positive conjunction that matters to the output.

The negative R^2 after boolean ablation also has a simple explanation. The boolean group contains complementary True and False coordinates. Setting both to zero creates a state that never occurs during normal execution. The readout

must extrapolate from this out-of-distribution state and performs worse than a constant mean predictor. The negative score describes the readout’s extrapolation after ablation.

The AB computation is additive in a basis that contains its completed detector, interactional in a basis that contains only its inputs, and additive again after those inputs are lifted with pair features. The writeback test shows that the recovered detector subspace is used by the final readout.

5.5 IOI replicates conditional backup and ranks the primary–negative interaction first

Tracr lets us check an observer against compiler labels and a known program. A pretrained language model removes that ground truth. We next examine the indirect-object identification (IOI) task in GPT-2-small. Prior work has identified a circuit for this task and documented several qualitative behaviors, including a conditional backup response [19, 20]. This gives us an intermediate case in which we know which groups to examine and roughly what they do, but lack the finite effect of every possible intervention.

Consider the prompt “When Alice and Bob went to the store, Alice gave a book to ...”. The model should predict Bob, the indirect object (IO), rather than Alice, the repeated subject (S). Three documented head groups contribute to this choice. Primary Name Movers write evidence for Bob toward the output. Backup Name Movers provide another route for writing Bob and become important when the primary route is weakened. Negative Name Movers write a negative signal against names that earlier components have already begun to copy. They act as a copy-suppression or calibration mechanism rather than as another answer-writing path [20]. On the IOI intervention surface studied here, their net effect lowers the IO-versus-subject readout, so removing them can improve that particular readout.

Let P denote the three primary Name Mover heads (9.9, 9.6, 10.0), B the eight Backup Name Movers (9.0, 9.7, 10.1, 10.2, 10.6, 10.10, 11.2, 11.9), and E the two Negative Name Movers (10.7, 11.10). A label such as 9.9 means layer 9, head 9. We measure the final-token margin

$$M = \text{logit}(\text{IO}) - \text{logit}(\text{S}).$$

In the example, $M > 0$ means that Bob ranks above Alice, while $M < 0$ means that Alice ranks above Bob. A change in M does not necessarily change the emitted token: the output flips between these two names only if the margin crosses zero, and another vocabulary token could also rank above both.

For an ablated set of heads U , let $\Delta(U)$ be the clean margin minus the ablated margin. Positive $\Delta(U)$ means that the

ablation moves the model away from Bob and toward Alice. Negative $\Delta(U)$ means that it moves the model toward Bob.

We use two studies because complete group ablations and prediction over ordinary subset masks answer different questions. The direct study uses 21 selected group masks, 128 matched name pairs, and eight prompt conditions formed by four lexical frames in the standard ABBA and BABA orders. These two orders reverse how the names are introduced relative to the person who later repeats as the subject. We repeat the study with two ablation conventions. Template-conditioned mean ablation replaces a selected head output with its average clean activation on that prompt template; zero ablation replaces it with zero. Agreement across the two reduces the chance that the interaction ordering comes from one replacement convention. This study asks what happens at the corners of the intervention space, where whole groups are removed.

The predictive study measures 240 masks over 256 name pairs on one prompt template. A masked head is replaced by its average clean activation on prompts from that template. We score the 239 non-clean masks using five cross-validation splits, while including the unablated mask in every training split to fix the reference response. This study asks which measurement model best predicts unseen subset interventions drawn from the tested mask distribution.

Direct group ablations reproduce conditional backup. For two groups A and B , define

$$I_{AB} = \Delta(A + B) - \Delta(A) - \Delta(B).$$

If their effects add independently, $I_{AB} = 0$. A positive value means that removing both groups causes more damage than we would predict by adding their separate effects.

The primary and backup groups give a concrete example. On the original template, removing P lowers the Bob–Alice margin by 0.463, while removing B lowers it by 0.350. An additive model therefore predicts that removing both groups will lower the margin by 0.813. The measured reduction is instead 1.869, giving $I_{PB} = 1.055$.

In terms of the prompt, the backup path looks relatively unimportant while the primary heads are still writing Bob. Once the primary path is gone, losing the backups becomes much more damaging. For illustration, suppose the clean model favors Bob by 1.5 logits. The additive estimate predicts that Bob will remain ahead by 0.687 after both groups are removed. The measured joint effect instead gives a margin of -0.369 , placing Alice ahead of Bob. The value 1.5 is illustrative, but it shows what the interaction means for an actual prediction: an additive observer can put the model on the wrong side of the Bob–Alice decision boundary.

The Negative Name Movers produce a different sign. Averaged over the eight combinations of lexical frame and name

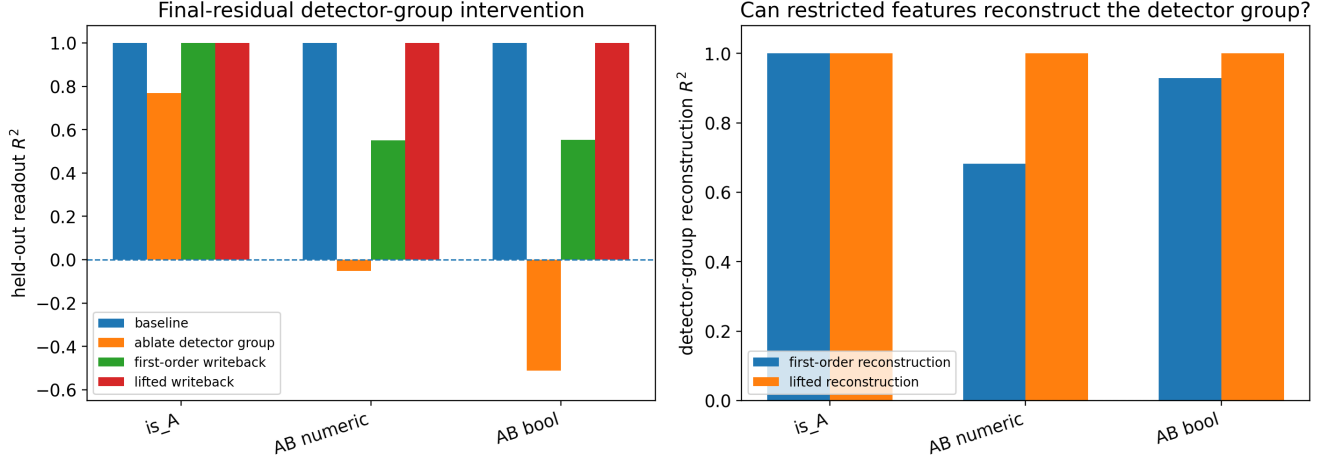


Figure 9: Detector-group writeback in Tracr circuits. Left: ablating the final-residual detector group destroys output prediction for the AB programs, while writing back the lifted reconstruction restores the original held-out R^2 . Right: additive features only partially reconstruct the detector groups, whereas lifted features reconstruct them completely.

Table 3: Direct IOI group interactions across four lexical frames in ABBA and BABA order, with 128 shared matched name pairs. Intervals use a crossed bootstrap that resamples lexical frames and one shared name-pair vector while retaining both orders. The last column divides by the number of possible cross-group head pairs; it does not assign the same effect to every individual pair.

Pair (heads)	mean ablation I [95%]	zero ablation I [95%]	I/pair (mean/zero)
$P \times B$ (24)	1.080 [0.912, 1.262]	1.268 [1.013, 1.558]	0.045/0.053
$P \times E$ (6)	1.830 [1.569, 2.076]	1.681 [1.474, 1.889]	0.305/0.28
$B \times E$ (16)	0.502 [0.429, 0.576]	0.518 [0.416, 0.618]	0.031/0.032

order, mean ablation gives $\Delta(E) = -2.270$ and $\Delta(P + E) = -0.136$; zero ablation gives -2.335 and -0.769 . Removing only E therefore improves the IO-versus-subject margin. This is consistent with copy suppression, since a mechanism that improves calibration over the broader model distribution can still oppose a narrow task-specific logit difference.

The primary-negative interaction is larger. The $P \times E$ result also has a direct interpretation in the Alice-Bob example. Under mean ablation, removing the primary heads lowers the Bob-Alice margin by about 0.304. Removing the Negative Name Movers while leaving the primary path intact raises the margin by 2.270. If these effects were independent, removing both groups would raise the margin by

$$2.270 - 0.304 = 1.966.$$

The observed increase is only 0.136.

The reason is conditional copy suppression. While the primary heads are writing Bob, the Negative Name Movers have a strong copied-name signal to act against. Removing the negative heads then releases that suppression and gives Bob a large boost. Once the primary heads have also been removed, much less Bob evidence remains for the negative heads to suppress. Removing them therefore provides almost no boost. The difference between the predicted and observed responses, 1.830 logits, is the measured $P \times E$ interaction.

Every tested combination of lexical frame and name order gives the same interaction ordering, $I_{PE} > I_{PB} > I_{BE}$, under both ablation conventions. The $I_{PE} - I_{PB}$ contrast is 0.750 [0.562, 0.934] under mean ablation and 0.413 [0.207, 0.611] under zero ablation. Both contrasts remain positive when we remove any one lexical frame.

The difference is not an artifact of group size. After dividing by the number of possible cross-group head pairs, the normalized $P \times E$ effect is 6.8 $\times$ the normalized $P \times B$ effect under mean ablation and 5.3 $\times$ under zero ablation.

The three group interactions capture part of the response, with a remaining three-way residual of 0.319 [0.215, 0.439] under mean ablation and 0.601 [0.417, 0.790] under zero ablation. Table 3 reports group-level effects; dividing by the number of cross-group head pairs gives a size-normalized comparison, while individual pair effects may differ.

Held-out subset prediction gives the same ordering. The predictive study tests whether these interactions help predict unseen subset masks beyond the complete group ablations. We stratify masks by the exact number of removed primary heads, a binned count of removed Backup Name Movers, and

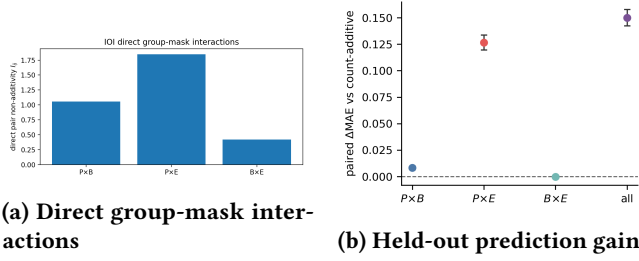


Figure 10: IOI interaction measurements. Left: direct group interactions on the original-template mean-ablation surface. Right: held-out gains from adding one count-product term to the count-additive model. $P \times E$ gives the largest of the three tested gains; the interval for $B \times E$ crosses zero.

the exact number of removed Negative Name Movers. We oversample masks with at least two removed primary heads because conditional backup is most visible after the primary path has been weakened.

A count-additive baseline gives each group its own flexible dose response. It can learn, for example, that removing four backup heads has more than twice the effect of removing two. What it cannot express is that removing a Negative Name Mover helps Bob while the primary writers remain but helps much less after those writers have been removed.

We represent that dependence by adding one normalized count product at a time:

$$q_P q_B, \quad q_P q_E, \quad q_B q_E,$$

where $q_P = n_P/3$, $q_B = n_B/8$, and $q_E = n_E/2$. Each term represents an interaction between group-level ablation counts; it does not identify a particular pair of heads.

The count-additive baseline has MAE 0.386 and $R^2 = 0.752$. An additive model with a separate effect for each head gives similar values: MAE 0.379 and $R^2 = 0.759$. Adding the PB , PE , or BE term gives MAE 0.377, 0.259, and 0.386, respectively, and $R^2 = 0.774$, 0.887, and 0.754.

Across 2,000 paired prompt bootstraps with fixed five-fold splits, PB improves MAE by 0.0083 (95% CI [0.0064, 0.0101]), while PE improves it by 0.1265 (95% CI [0.1195, 0.1338]). The BE change is -0.0002 (95% CI [-0.0015 , 0.0011]), which is indistinguishable from no improvement. Adding all three terms gives MAE 0.236, $R^2 = 0.907$, and an MAE improvement of 0.1499 (95% CI [0.1425, 0.1578]).

The ordering remains stable across 200 alternative assignments of intervention masks to cross-validation folds. The PE gain is positive in all 200 assignments, PB in 199, and BE in 154. Among the three tested single interaction terms, PE is both the largest and the most stable. PB gives a smaller but repeatable gain, while the BE gain depends on the split.

What mechanistic tomography adds. Prior IOI work supplies the candidate head groups and their qualitative roles. Mechanistic tomography supplies the procedure for testing whether those groups are sufficient for the intended prediction task. We first fit the additive family and test it on held-out interventions. Its residual shows that separate group effects cannot predict the finite response, so we lift the design with the three group-pair terms. The $P \times E$ column expresses that removing Negative Name Movers has one effect while the primary path remains and another after it has been weakened; adding it reduces held-out MAE from 0.386 to 0.259. Using these documented groups, MT identifies when the additive family needs additional structure and which relations improve finite-intervention prediction under the declared distribution.

What the two measurements tell us. The direct and predictive studies agree on the ordering of the three tested interactions while measuring different aspects of their effects. A complete $P + B$ ablation measures the corner where both groups are gone. A predictive term measures how useful that interaction is across the subset masks the observer will encounter. A strong corner interaction can make only a small average predictive contribution if the measurement distribution rarely approaches that corner.

This distinction explains why the known $P \times B$ backup response is clear under complete group ablation yet contributes much less to prediction than $P \times E$ over the tested subset distribution. For that distribution, one prompt template, and mean ablation, $P \times E$ is the strongest and most stable of the three group-level terms. The broader direct-ablation study establishes that the group ordering also holds across the tested prompt conditions and both replacement conventions.

5.6 Finite calibration is sufficient on the tested Qwen-2.5-7B surface

The preceding experiments use exact posteriors, planted effects, compiler labels, or documented circuit groups. Qwen tests the same measurement procedure against a behavioral response surface. Here the model’s measured response to a finite intervention supplies the reference for evaluating predictive transfer; an independently known internal mechanism is unavailable.

We fix Qwen-2.5-7B-Instruct [27] and a decline-versus-compliance margin. The readout compares the model’s log probability of four decline stems with its log probability of four compliance stems at the first generated position. Harmful prompts come from HarmBench, and safe, superficially safety-related prompts come from XSTest [22, 28]. Together they provide a varied response surface for testing intervention prediction. The endpoint is computed directly from model probabilities, without generating a continuation

Table 4: Held-out prediction on the Qwen-2.5-7B decline-versus-compliance response surface. The test contains 128 actions at the unseen scale 0.75 and 224 prompts.

Response map	parameters	MAE	RMSE	R^2
Calibrated additive	20	0.003790	0.004699	0.9829
Lifted pairwise	48	0.003801	0.004613	0.9835

or using an LLM judge, and measures this margin rather than overall model safety.

The intervention basis contains eight layerwise harmful-versus-benign contrast directions, measured at layers 5, 8, 11, 14, 17, 20, 23, and 26 at the last prompt token. An *action* is a vector of signed steering weights over these directions, normalized to the chosen intervention scale. We hold the basis, prompt splits, and action library fixed. The 401-action library combines random masks at four densities with unablated and single-direction controls. Calibration and validation use scales 0.5 and 1.0; the final test crosses 128 held-out actions at scale 0.75 with 224 held-out prompts.

We compare two response maps. The calibrated additive map contains the declared first-order action features and scale corrections. The lifted map adds all 28 products between the eight layer coefficients. Both models choose their ridge penalty on the same validation split. We then evaluate both on the same held-out action-prompt pairs.

The additive map gives held-out $R^2 = 0.9829$ and MAE 0.003790. The lifted map gives $R^2 = 0.9835$ and MAE 0.003801. The estimated relative MAE improvement from lifting is -0.29% . A paired prompt-family-by-action bootstrap gives a 95% interval of $[-3.56\%, 5.65\%]$.

We retain the calibrated additive map for this basis, prompt distribution, and intervention regime because adding pair terms yields no detected MAE improvement. The interval spans zero and extends slightly beyond the predeclared 5% practical threshold, so we stop at the available precision; the comparison does not establish equivalence.

This result complements the IOI study. In IOI, the additive residual points to a specific missing relation, and the $P \times E$ term materially improves held-out prediction. On the Qwen behavioral surface, finite calibration makes the simpler map adequate at the measured precision.

The public artifact repository contains the complete experiment, frozen measurements, hashes of the source code and software environment, and a Colab notebook that regenerates Table 4 without loading the model. A separate GPU script can rerun the finite measurements from the pinned model revision.

6 Related work

Network tomography and measurement design. Network tomography infers hidden quantities from aggregate observations [11, 36], combining direct measurement with inference [45] and allocating probes through Fisher information [46]. MT treats the neural network itself as the system being measured and studies intervention masks through approximation error and conditioning. Appendix E connects finite interventions to the classical balance of variance and model bias [47]; Fisher-information-optimal allocation and online guarantees remain open extensions.

Measuring intervention effects. Patching and causal tracing measure finite effects [23, 38], while attribution patching uses gradients [32]. Integrated Hessians and HVP correction account for curvature [14, 39]. Coalition methods assign main and interaction effects [18, 35], while compressed-sensing methods recover sparse effects from aggregate masks [4, 7, 15]. MT asks which measurements predict the intended finite interventions. This differs from linear artificial tomography in representation engineering, which estimates concept directions from contrastive activations [37, 40].

Ground truth and causal validation. Bayesian testbeds expose learned representations by training transformers on systems with known posteriors [1, 30]. Our exact HMM posterior provides an external reference for testing estimation and control. Causal abstraction and DAS test higher-level variables [12, 13], while MIB compares targets, ablations, and causal metrics [24]. MT complements these approaches by testing effect-map recovery, then evaluating the map through writeback and control.

Decisions and control. Steering evaluations test interventions and their reliability [6, 33, 37], while actionable interpretability advocates judging insights by the decisions they support [26]. PID Steering provides feedback control [25], while MERA selects and calibrates interventions with an option to abstain [42]. MT focuses on how measurement errors affect a fixed controller, following the distinction between reconstruction accuracy and routing quality in network tomography [29]. Decision-aware experimental design chooses measurements directly for downstream utility [43, 44]. MT’s control tests evaluate an estimate’s usefulness for a fixed controller.

7 From effect maps to tested observers

Mechanistic tomography gives researchers a workflow rather than a ranking of interpretability methods. State the target and operating region, choose the least costly measurement family allowed by the available access, test it on held-out interventions, and use the remaining error to decide what to measure next. If the resulting map will guide an intervention, test it through that action as well.

Table 5: Observer selection as a measurement decision.

Regime	Available access	Start with	Escalate when
Additive, white-box	gradients and finite evaluations	AtP, then a few finite probes	held-out error does not reduce to a small gain correction
Additive, forward-only	scalar intervention responses	sparse subset measurements	the map is dense or held-out prediction fails
Interactional, white-box	gradients, HVP queries, and finite evaluations	AtP plus HVPs for local curvature	local support fails, or finite-scale probes show that calibration is needed
Interactional, forward-only	scalar intervention responses	lifted subset measurements	lifted coverage or conditioning fails
Uncertain basis	candidate coordinates and interventions	simplest declared basis	residuals persist or writeback fails
Downstream use	fixed controller and actuator	lowest-dimensional validated observer	target, collateral, or transfer tests fail

Table 5 summarizes the decision path. For example, when gradients are available, AtP supplies the local map directly, saving the forward evaluations needed to reconstruct it. Spend the finite evaluations on testing and calibration, and move to HVPs or lifted measurements only when the first-order family leaves structured held-out error.

State the intended use before choosing the measurements. The methods in Table 1 do not estimate the same object. A study should state its target quantity, component basis, model access, intervention scale, mask distribution, context distribution, and intended use. A gradient may explain which direction initially raises a readout, while an observer must predict what happens after a finite action in that direction.

Start with the cheapest measurement family that can represent the target. Cheapest means cheapest under the available access regime. With forward-only access, aggregate masks can recover a compressible finite-effect map with fewer evaluations than probing every component. With gradients, one backward pass already supplies the local map, so finite evaluations should test that map rather than reconstruct it. HVPs and lifted masks become useful when the first-order family cannot represent the response.

Let held-out residuals choose the next measurement. A poor fit by itself does not say what to do next; the structure of the held-out error does. A shared scale error suggests one gain, group-specific errors suggest a larger calibration family, and conditional effects require interactions. If a completed detector makes an apparently interactional computation additive, the basis was the limiting choice. In IOI, the $P \times E$ column supplies exactly the dependence the additive model cannot express (Sec. 5.5). More measurements of an inadequate family would only estimate the wrong response more precisely.

Validate an observer through its intended action. Reconstruction and downstream performance answer different questions. They usually agree, as the HMM error correlation shows, but the actuator weights errors unevenly. When an estimate guides an intervention, first test whether it reconstructs or predicts the declared quantity. Then hold the controller and actuator fixed and test the action. The nuisance-state experiment shows why target performance must be considered alongside unwanted internal movement. Reconstruction, intervention prediction, and control performance are related but distinct tests.

The same held-out error can arise because the intervention is too large for a local map, the response contains an omitted pair effect, or the coordinates omit the required detector. The ground-truth progression in Sec. 3 separates these causes; each level supplies the external reference the diagnosis requires.

Each study makes a different part of the measurement problem observable. The planted interaction experiment fixes a sparse support and a basis containing its pair terms; Tracr provides compiler labels; IOI provides documented groups whose direct ordering is stable across eight combinations of lexical frame and name order and two ablation conventions. IOI’s predictive comparison uses one prompt template and mean ablation, while Qwen’s decline-versus-compliance margin tests behavioral prediction in a larger instruction-tuned model. Together these studies establish concrete operating regimes for the procedure. Testing its prevalence and scaling requires additional models and bases, and applying the scale–density criterion requires a measured $\epsilon \times \rho$ response surface.

Current mechanistic interpretability benefits from a strong architectural prior, since researchers know where to look because people chose the layers, attention heads, residual streams, and token positions. If models increasingly help design their successors, those conventions may become less transparent to the people evaluating them. This strengthens the case for a measurement-first approach whose claims rest on declared interventions and held-out responses without requiring a familiar blueprint.

ObserverBench provides a complementary workbench for comparing this decision path across models, tasks, observers, and interventions [10]. Tasks declare the controlled model or system, intervention distribution, controller, actuator, and held-out metrics, while researchers supply the observer. Holding these choices fixed makes differences between observers comparable across case studies.

A shared interface lets researchers compare attribution maps, SAE features, learned probes, interaction-aware estimators, and new designs through the same prediction, control, collateral, and robustness measures. Mechanistic tomography supplies the logic for deciding what must be measured

and validated; ObserverBench tests how well different observers satisfy that contract as the available ground truth becomes weaker.

Across the tested settings, the procedure recovered a compressible map from few aggregate measurements, calibrated gradients with a handful of finite probes, identified the dominant predictive group interaction in GPT-2, and stopped at the calibrated additive map on Qwen. These results turn the common inverse formulation into a practical measurement workflow: acquire the information the intended intervention needs, and use held-out response error to decide when to refine the estimate and when to stop.

AI assistance. Generative AI tools assisted with writing experimental code, editing the manuscript text, and developing the bias-aware design derivations in Appendix E. The author is responsible for the methods, results, and claims reported here.

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A From one finite intervention to the tomography equation

The model is nonlinear, but the tomography equation does not require it to be globally linear. It requires a more limited statement: around each clean activation, the response to a finite intervention can be separated into a local linear contribution and a residual. The proof performs that separation for one mask and then stacks the resulting measurements.

PROOF. Step 1: Treat the mask as one activation-space direction.

Fix a context c . The mask a assigns a coefficient to each component direction, so the complete activation displacement is

$$u_c = D_c a. \quad (16)$$

This notation turns an intervention that may touch many components into one direction u_c through activation space. The scalar ε determines how far the intervention moves along that direction.

Step 2: Separate the tangent response from curvature.

Apply the second-order Taylor theorem along the path from h_c to $h_c + \varepsilon u_c$. There is a point ξ_c on this segment such that

$$\begin{aligned} f_c(h_c + \varepsilon u_c) - f_c(h_c) &= \varepsilon \nabla f_c(h_c)^\top u_c \\ &\quad + \frac{\varepsilon^2}{2} u_c^\top \nabla^2 f_c(\xi_c) u_c. \end{aligned} \quad (17)$$

The first term is the response predicted by the tangent at the clean activation. The second is the part caused by curvature along the finite path. This is the only place where differentiability enters the reduction.

Step 3: Express the tangent response as a measurement of component effects.

Because $D_c = [d_1(c), \dots, d_n(c)]$,

$$\nabla f_c(h_c)^\top D_c a = \sum_{i=1}^n a_i \nabla f_c(h_c)^\top d_i(c). \quad (18)$$

Each term is the local effect of one component direction, multiplied by the coefficient assigned to it by the mask.

The target map is

$$\begin{aligned} x_i &= \mathbb{E}_c \left[\frac{\partial}{\partial \delta} f_c(h_c + \delta d_i(c)) \Big|_{\delta=0} \right] \\ &= \mathbb{E}_c [\nabla f_c(h_c)^\top d_i(c)]. \end{aligned} \quad (19)$$

Averaging the tangent response over contexts therefore gives

$$\mathbb{E}_c [\nabla f_c(h_c)^\top D_c a] = a^\top x. \quad (20)$$

This is the key reduction. The finite intervention may change many components, but its first-order response is one linear measurement of their unknown effects. The mask supplies the weights in that measurement.

Divide the Taylor expansion by ε and average over contexts:

$$y_\infty(a, \varepsilon) = a^\top x + r(a, \varepsilon), \quad (21)$$

where

$$r(a, \varepsilon) = \frac{\varepsilon}{2} \mathbb{E}_c [u_c^\top \nabla^2 f_c(\xi_c) u_c]. \quad (22)$$

The division by ε removes the intervention scale from the linear term. One power of ε remains in the curvature term, which is why nonlinear error grows linearly with the intervention scale after normalization.

Using $\|\nabla^2 f_c\|_{op} \leq L$,

$$\begin{aligned} |r(a, \varepsilon)| &\leq \frac{\varepsilon}{2} \mathbb{E}_c [\|\nabla^2 f_c(\xi_c)\|_{op} \|u_c\|_2^2] \\ &\leq \frac{L}{2} \varepsilon \mathbb{E}_c \|D_c a\|_2^2. \end{aligned} \quad (23)$$

The bound depends on both the scale ε and the displacement produced by the mask. A dense mask need not have the same nonlinear error as a sparse mask, even when both use the same scalar intervention size.

Step 4: Add the error from estimating the response.

The population response is not normally available exactly. We estimate it from a finite batch and possibly from stochastic or finite-precision model evaluations. Write the empirical response as

$$\tilde{y}_B(a, \varepsilon) = y_\infty(a, \varepsilon) + \frac{e_B(a)}{\varepsilon} + \zeta_B(a, \varepsilon). \quad (24)$$

The two empirical terms appear separately because they scale differently. The quantity $e_B(a)$ is error in the unnormalized response numerator. After we divide the response by ε , this error becomes $e_B(a)/\varepsilon$. It can therefore dominate when the intervention is very small.

The term $\zeta_B(a, \varepsilon)$ represents sampling error that remains after normalization. Pairing the clean and intervened evaluations can cancel much of their shared variation. This term therefore need not grow as $1/\varepsilon$. Pairing is part of the measurement design, not only a statistical convenience.

Combining the curvature and empirical terms gives

$$\tilde{y}_B(a, \varepsilon) = a^\top x + w(a, \varepsilon), \quad (25)$$

with

$$w(a, \varepsilon) = r(a, \varepsilon) + \frac{e_B(a)}{\varepsilon} + \zeta_B(a, \varepsilon). \quad (26)$$

Hence

$$|w(a, \varepsilon)| \leq \frac{L}{2} \varepsilon \mathbb{E}_c \|D_c a\|_2^2 + \frac{\tau_B(a)}{\varepsilon} + \nu_B(a). \quad (27)$$

Step 5: Stack the measurements.

Repeat the construction for masks $a_1, \dots, a_m$. Put each row $a_j^\top$ into the matrix A , each empirical response into $\tilde{y}$, and each residual into w . The m scalar equations then become

$$\tilde{y} = Ax + w. \quad (28)$$

□

The proof establishes the observation model used by mechanistic tomography. It says that each finite intervention supplies a linear equation about the local component-effect map, with a residual whose dependence on intervention scale and mask displacement is explicit.

Identifying the map additionally requires rank, coverage, or sparse-recovery properties of A . Because curvature and sampling error can depend on the mask, the residual remains design-dependent. The next appendix connects these recovery conditions to the measurement model and explains how gradient access changes the procedure.

B When the measurements identify the effect map

The measurement equation from Appendix A permits recovery only when the design separates the possible effect maps. Identical columns, for example, reveal a combined effect but cannot distinguish its components. The relevant conditions depend on whether the target is unrestricted or sparse, and on whether its coordinates represent main effects or interactions.

Sparse maps need fewer well-designed measurements. For $\tilde{y} = Ax + w$ and a residual bound $\eta \geq \|w\|_2$, basis-pursuit denoising minimizes $\|z\|_1$ subject to $\|Az - \tilde{y}\|_2 \leq \eta$. Under the classical restricted-isometry conditions, its estimate satisfies [8]

$$\|\hat{x} - x\|_2 \leq C_0\eta + C_1 \frac{\sigma_k(x)_1}{\sqrt{k}}, \quad (29)$$

where $\sigma_k(x)_1$ is the ℓ_1 error of the best k -sparse approximation. The first term propagates finite-response, sampling, and basis error; the second accounts for approximate sparsity. Suitably normalized sub-Gaussian designs provide the required conditioning with high probability at $m \gtrsim k \log(n/k)$; OMP has its own incoherence or restricted-isometry guarantees [34]. These are classical recovery results applied to MT's intervention-derived equation.

Why coordinate patching has a forward-only lower bound. Coordinate patching uses rows that select one coordinate at a time. With fewer than n such measurements, at least one coordinate remains unobserved. The zero map and a map supported only on that unobserved coordinate then produce exactly the same observations. No recovery rule can distinguish them in the worst case.

This is a worst-case separation result, not a claim that every real transformer requires all n patches. A prior over likely supports can improve average or Bayesian performance. The limitation also applies specifically to forward-only measurements. With white-box access, one backward pass returns every coordinate of the local first-order map directly. In the language of mechanistic tomography, attribution patching

is therefore the direct-access first-order case: the measurement procedure returns x without requiring an inverse solve. Finite probes remain useful because they test whether that local map predicts the response at the scale where it will be used.

Pair interactions become coordinates after lifting. An additive map is insufficient when the effect of one component depends on whether another component is also changed. Proposition 3 handles this by treating each pair as an additional unknown coordinate.

Define the finite-scale parameter

$$\theta_\varepsilon = (x_\varepsilon, \text{vec } \Gamma_\varepsilon) \in \mathbb{R}^N \quad (30)$$

and replace each original design row a by

$$\phi(a) = [a_1, \dots, a_n, \{a_i a_j\}_{i < j}]. \quad (31)$$

A response containing main and pair effects can then be written as

$$\begin{aligned} \tilde{y}(a) &= a^\top x_\varepsilon + \sum_{i < j} a_i a_j \Gamma_{\varepsilon, ij} + w_\varepsilon(a) \\ &= \phi(a)^\top \theta_\varepsilon + w_\varepsilon(a). \end{aligned} \quad (32)$$

Stacking rows yields

$$\tilde{y} = \Phi(A)\theta_\varepsilon + w_\varepsilon. \quad (33)$$

For two components, the model reads

$$\tilde{y}(a) = a_1 x_1 + a_2 x_2 + a_1 a_2 \Gamma_{\varepsilon, 12} + w_\varepsilon(a).$$

Measurements that change only the first component reveal x_1 ; measurements that change only the second reveal x_2 . The additional response that appears when both are changed reveals the pair term $\Gamma_{\varepsilon, 12}$. Lifting turns that interaction into an ordinary column of a larger linear inverse problem.

If θ_ε is sparse or compressible and the lifted design $\Phi(A)$ separates sparse alternatives, the preceding stable-recovery result applies with A replaced by $\Phi(A)$ and x replaced by θ_ε . This establishes Proposition 3.

The condition on the lifted design is essential. A design that separates main effects need not separate pair effects. If two components always occur together, for example, the pair column $a_i a_j$ may duplicate a main-effect column. Different pairs can also produce identical columns under sparse or constrained masks. Collecting more measurements with the same aliased pattern does not solve the problem; the intervention design must change.

This is why held-out and explicitly de-aliased masks matter. They test whether a recovered pair represents an interaction that transfers beyond the masks used to fit it, rather than an arbitrary explanation of correlated columns.

What this appendix establishes. The recovery theorem used here is classical. The contribution is the reduction that connects finite mechanistic interventions to that theorem and makes the access assumptions explicit. Under forward-only access, aggregate masks can replace exhaustive coordinate probes when the declared map is sparse and the design separates sparse alternatives. With gradients, the first-order inverse problem disappears, but finite measurements are still needed to assess the residual at the intended intervention scale. When additive measurements fail, lifting shows exactly what must be added to measure pair interactions.

The formulation separates three properties that an experiment must check: sparsity concerns the chosen target, conditioning concerns the measurement design, and persistent held-out error concerns the adequacy of the measurement family. Diagnosing them separately identifies whether the next step should change the basis, improve measurement coverage, or add interaction terms.

C Why interaction measurements need a new design check

Lifting makes interactions visible by adding a column $a_i a_j$ for every candidate pair. It also creates a new identification problem. A mask matrix that separates the main-effect columns a_i may fail to separate the pair columns, so good first-order conditioning does not imply good conditioning after lifting.

A three-component half-fraction design separates the two issues. Use the four signed masks satisfying $a_1 a_2 a_3 = 1$:

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix}, \quad A^\top A = 4I_3.$$

The main-effect columns are orthogonal, so $A/2$ preserves the norm of every main-effect vector. Yet on every row,

$$a_1 a_2 = a_3, \quad a_1 a_3 = a_2, \quad a_2 a_3 = a_1.$$

In the lifted column order $(a_1, a_2, a_3, a_1 a_2, a_1 a_3, a_2 a_3)$, the first and sixth columns are identical. A unit main effect on component 1 and a unit pair effect on components 2 and 3 therefore give exactly the same measurements. The lifted design fails to distinguish even these two one-sparse maps, despite perfect conditioning of the main-effect design. Repeating rows from this half-fraction preserves the alias; masks from the other half, where $a_1 a_2 a_3 = -1$, distinguish the two explanations.

Dense independent Rademacher masks provide a useful ideal reference. Their entries take independent values in $\{-1, +1\}$, making degree-one columns a_i and degree-two columns $a_i a_j$ orthogonal in the population. A finite experiment only approximates that property. Real intervention families may also need sparse masks, fixed mask sizes, or

exclusions that preserve model behavior. Those constraints can make pair columns highly correlated or identical.

The lifted design must therefore be checked directly. Its singular values, restricted conditioning, and column correlations answer whether the proposed masks can distinguish the candidate pairs. A separate set of masks should also change the co-occurrence patterns that appeared during fitting. If the recovered pair predicts those held-out responses, it reflects an interaction that transfers beyond a training alias.

This check explains why interaction-aware tomography is more than adding pair terms to a regression. The new columns require new measurement coverage. Sec. 5.3 uses planted supports to verify this logic under ground truth, and Sec. 5.5 applies it to documented head groups in IOI, where held-out interventions provide the available test.

D Proofs for calibration, scale choice, interaction blindness, and HVP measurements

The four results proved here answer successive measurement-design questions. Proposition 1 asks when a small correction can repair a baseline observer. Corollary 1 asks how intervention scale and mask density affect its error. Proposition 2 shows why no first-order method can recover a pure pair interaction. Corollary 2 then explains what white-box curvature measurements recover and when they predict finite interventions.

PROOF OF PROPOSITION 1. The proposition states when exact calibration is identifiable and bounds the approximation error for every correction family. When the finite response lies in the family, the bound is zero; a positive bound measures the error that calibration cannot remove.

Step 1: Express the calibration measurements as a small linear system.

Let $a_1, \dots, a_m$ be the calibration masks. Put the feature row $\phi(a_j)^\top$ into row j of Φ , and define

$$M = \Phi B.$$

The matrix B contains the r allowed correction directions. The matrix M therefore records how those corrections appear under the chosen calibration masks.

If the finite response belongs to the declared family, then for some θ_ϵ ,

$$F_\epsilon(a_j) = \phi(a_j)^\top (x_0 + B\theta_\epsilon) \quad \text{for every } j. \quad (34)$$

After subtracting the fixed baseline response Φx_0 , the measurements satisfy

$$y - \Phi x_0 = M\theta_\epsilon + v, \quad (35)$$

where v contains finite-sample and measurement error.

This is a smaller inverse problem than estimating the complete effect map. The unknown has dimension r , not q .

Step 2: Use rank to establish identifiability.

In the noiseless case, suppose two corrections θ_1 and θ_2 produce the same measurements. Then

$$M(\theta_1 - \theta_2) = 0.$$

If M has full column rank, its null space contains only zero. Hence $\theta_1 = \theta_2$, and the correction is unique.

An r -dimensional correction therefore needs only r independent equations in the ideal case. The statement uses $O(r)$ because practical experiments normally collect additional measurements to improve conditioning and to reserve separate masks for held-out validation.

Step 3: Show how measurement error propagates into the correction.

When M has full column rank, the least-squares estimate obeys

$$\widehat{\theta} - \theta_\varepsilon = M^\dagger v, \quad (36)$$

where $M^\dagger$ is the pseudoinverse. Therefore

$$\|\widehat{\theta} - \theta_\varepsilon\|_2 \leq \|M^\dagger\|_2 \|v\|_2. \quad (37)$$

The factor $\|M^\dagger\|_2$ is the price of a poorly conditioned calibration design. Full rank makes the correction identifiable; good conditioning makes it stable.

Step 4: Identify the error that calibration cannot remove.

For any finite-response function, every observer in the declared affine family has the form

$$a \mapsto \phi(a)^\top (x_0 + B\theta)$$

for some θ . Its held-out error under Q is therefore at least

$$\inf_{\theta} \|F_\varepsilon(\cdot) - \phi(\cdot)^\top (x_0 + B\theta)\|_{L_2(Q)}. \quad (38)$$

This lower bound holds even with infinitely many noiseless calibration measurements. It is zero when the family represents the response under Q . A positive value measures approximation error that more calibration data cannot remove.

The calibration dimension is thus the smallest declared correction dimension whose held-out residual falls within the chosen tolerance. $\square$

The proof gives calibration dimension a direct interpretation. A low dimension means that the baseline observer has the right structure and needs only a few finite-scale corrections. A persistent residual means something stronger: the observer family is missing a feature, such as an interaction, and must be enlarged.

PROOF OF COROLLARY 1. Two effects matter. The intervention creates measurement error, and the inverse solve may

amplify it. We first bound the former and then pass it through the latter.

Step 1: Bound the nonlinear error of one mask.

Lemma 1 bounds the curvature contribution for a normalized mask a by

$$\frac{L}{2} \varepsilon \mathbb{E}_c \|D_c a\|_2^2. \quad (39)$$

For the mask family $\mathcal{A}_\rho$, the definition

$$\kappa_\rho^2 = \sup_{a \in \mathcal{A}_\rho} \mathbb{E}_c \|D_c a\|_2^2$$

therefore gives the uniform bound

$$\frac{L}{2} \varepsilon \kappa_\rho^2. \quad (40)$$

This term grows with ε because a larger intervention travels farther from the local linear approximation.

Step 2: Add the two sampling terms.

The same lemma separates error already present in the response numerator from normalized sampling error. Uniformly over $\mathcal{A}_\rho$, the complete first-order measurement bound is

$$\eta_1(\varepsilon, \rho) = \frac{L}{2} \varepsilon \kappa_\rho^2 + \frac{\tau_B(\rho)}{\varepsilon} + v_B(\rho). \quad (41)$$

The first term grows with scale. The second grows when scale becomes too small. The third does not have to vary with scale after normalization.

Step 3: Pass measurement error through the recovery method.

The per-mask bound gives $\|w\|_\infty \leq \eta_1(\varepsilon, \rho)$. The assumed stability inequality converts this maximum normalized measurement residual into ℓ_2 reconstruction error. If an available theorem instead gives $\|\widehat{x} - x\|_2 \leq K(A_\rho) \|w\|_2$ for m measurements, one may take $C_{\text{rec}}(A_\rho) = \sqrt{m} K(A_\rho)$. Any separate approximation term for a compressible target must be added to the bound. For the declared target class, the resulting measurement-induced recovery-error bound is

$$\begin{aligned} \mathcal{J}_1(\varepsilon, \rho) &= C_{\text{rec}}(A_\rho) \eta_1(\varepsilon, \rho) \\ &= C_{\text{rec}}(A_\rho) \left[\frac{L}{2} \varepsilon \kappa_\rho^2 + \frac{\tau_B(\rho)}{\varepsilon} + v_B(\rho) \right]. \end{aligned} \quad (42)$$

This multiplication connects the model side of the experiment to the inverse-problem side. Even a small response error can produce a poor estimate when the design is badly conditioned.

Step 4: Choose the scale for a fixed density.

Fix ρ . Assume $\tau_B(\rho) > 0$ and $L\kappa_\rho^2 > 0$, and treat $C_{\text{rec}}(A_\rho)$ and $v_B(\rho)$ as locally independent of ε . The scale-dependent part is

$$\frac{L}{2} \varepsilon \kappa_\rho^2 + \frac{\tau_B(\rho)}{\varepsilon}.$$

It is convex for $\varepsilon > 0$. Differentiating gives

$$\frac{L}{2}\kappa_\rho^2 - \frac{\tau_B(\rho)}{\varepsilon^2}. \quad (43)$$

Setting this derivative to zero yields

$$\varepsilon^*(\rho) = \sqrt{\frac{2\tau_B(\rho)}{L\kappa_\rho^2}}. \quad (44)$$

At this scale, the bound balances the cost of moving too far from the tangent against the cost of dividing numerator error by an intervention that is too small. The interior solution applies when it lies within the feasible scale range; otherwise the bound is minimized at a boundary of that range. If $L\kappa_\rho^2 = 0$ or $\tau_B(\rho) = 0$, the corresponding scale-dependent term is absent and this interior balance does not apply.

Step 5: Retain density as a design variable.

The calculation above optimizes scale only after fixing ρ . Density still changes κ_ρ , the coverage of the unknown coordinates, and the conditioning factor $C_{\text{rec}}(A_\rho)$. The joint design therefore minimizes $\mathcal{J}_1(\varepsilon, \rho)$ over both variables rather than choosing density independently. $\square$

This is a conditional design rule, not a universal numerical prescription. The preferred scale depends on curvature and noise, while the preferred density also depends on the actuator normalization and the conditioning of the mask matrix.

Second-order targets require a different scale calculation. The first-order criterion cannot be copied unchanged when the target is a Hessian or a pair-effect map. A local curvature estimate subtracts the first-order response and divides the remaining response by ε^2 . Its higher-order remainder and numerator error therefore scale differently from the terms above. If the target is instead the finite pair effect at a particular intervention size, then ε helps define the target itself and should match the scale at which the estimate will be used. In both cases, mask density still determines whether the relevant pairs receive independent coverage and whether the lifted design is well conditioned.

PROOF OF PROPOSITION 2. The proposition constructs a term that is invisible to every first-order observation but visible when the two relevant components are changed together.

Step 1: Record what a first-order method sees.

By assumption,

$$x_i = x_j = 0.$$

A gradient or singleton first-order measurement therefore reports zero for both components. These observations cannot distinguish a model with $H_{0,ij} = 0$ from one with $H_{0,ij} \neq 0$.

Step 2: Change both components in one intervention.

Let a mask assign coefficients a_i and a_j to the two directions. The third-derivative bound makes the second-order

Taylor remainder $O(\varepsilon^3)$ for a fixed mask. Specifically, if $\|\nabla^3 f\|_{op} \leq L_3$ and the activation displacement is εu , the remainder has magnitude at most $L_3 \varepsilon^3 \|u\|_2^3/6$. The expansion contains the cross contribution

$$\frac{1}{2}\varepsilon^2 (2a_i a_j H_{0,ij}) + O(\varepsilon^3) \quad (45)$$

before normalization. The quadratic cross term is zero when either component is absent and nonzero when the mask activates both; higher-order terms remain in the residual.

Step 3: Express the cross term as a lifted measurement.

Dividing the finite response by ε gives

$$a_i a_j \Gamma_{\varepsilon,ij}, \quad \Gamma_{\varepsilon,ij} = \varepsilon H_{0,ij} + O(\varepsilon^2). \quad (46)$$

The product $a_i a_j$ is therefore the measurement column associated with the pair. In a quadratic response model, there are no higher-order terms and the relation is exact.

A lifted design that separates this pair column from the other main and pair columns can recover the interaction. Every purely first-order design misses it because the required column is absent. $\square$

The issue is representational, not algorithmic. More gradient measurements cannot reveal a pure cross term when the first-order coordinates are zero. The measurement family must include a feature that changes only when the two components occur together.

PROOF OF COROLLARY 2. The corollary asks what white-box access contributes after a gradient has already supplied the complete first-order map.

Step 1: Separate gradient and curvature information.

A backward pass returns

$$\nabla f(h),$$

which determines the local first-order effects. It does not return the entries of the Hessian H_0 , which describe how those effects change as the activation moves.

Step 2: Use HVPs as measurements of curvature.

For a chosen direction v , a Hessian-vector product returns

$$H_0 v.$$

This is linear in the unknown Hessian. Choosing directions $v_1, \dots, v_m$ therefore produces designed linear measurements of the local curvature map. Selected coordinates or projections of these vectors can be stacked into an inverse problem in the same way as the forward measurements used for the first-order map.

Step 3: Recover a structured local pair map.

If the relevant entries of H_0 are sparse or compressible and the HVP design separates the possible supports, standard sparse recovery applies. The result is a local curvature map: it describes pair response at the operating point.

Step 4: Determine when local curvature predicts a finite intervention.

If the response is quadratic along the intervention path, the Hessian is constant and

$$\Gamma_\varepsilon = \varepsilon H_0.$$

Multiplying the recovered curvature map by ε then gives the normalized finite pair map exactly.

For a general nonlinear path, the Hessian can change as the activation moves. A single HVP at the starting point then identifies only local curvature. Predicting the finite response requires integrating curvature along the path or calibrating the local estimate against finite interventions. $\square$

This distinction mirrors the first-order case. White-box access provides the local measurement directly, but it does not remove the finite-scale question. HVPs recover local curvature; lifted forward measurements recover the pair response at the scale at which they are taken.

What the four proofs establish. Together, the proofs make the decision sequence precise. A small calibration is sufficient only when the missing response lies inside the declared correction family. Scale and density must be chosen with both nonlinear error and recovery conditioning in view. Pure interactions require pair-aware measurements. HVPs provide an efficient white-box route to local pair structure, but finite use still requires a quadratic response, path integration, or finite calibration.

E A bias-aware criterion for finite measurements

A measurement can reduce sampling variance while increasing approximation error. This appendix makes that tradeoff explicit for full-rank calibration and small effect maps. It specializes classical bias-aware design [47] using the intervention error in Lemma 1. The estimator, target, noise model, and measurement costs remain fixed when designs are compared. Full column rank distinguishes this setting from the underdetermined sparse fits elsewhere in the paper.

E.1 From measurement error to estimation risk

Fix a design S before observing its noise. Write

$$y = A\theta + b + \eta, \quad \mathbb{E}\eta = 0, \quad \mathbb{E}[\eta\eta^\top] = \Sigma \succ 0, \quad (47)$$

where $A \in \mathbb{R}^{m \times d}$ has full column rank, θ is the target, b is deterministic approximation or systematic error, and η is centered sampling noise. Correlated measurements are allowed. Let $W \succeq 0$ specify the same target loss for every design, with $\|z\|_W^2 = z^\top W z$. Define

$$M = A^\top \Sigma^{-1} A, \quad G = M^{-1} A^\top \Sigma^{-1}, \quad \hat{\theta} = Gy. \quad (48)$$

Here $\hat{\theta}$ is the generalized least-squares (GLS) estimator. The identity below uses the true covariance; a fitted covariance would require accounting for its estimation error.

PROPOSITION 4 (FIXED-DESIGN RISK). *Under (47)–(48),*

$$\mathbb{E}\|\hat{\theta} - \theta\|_W^2 = \text{tr}(WM^{-1}) + \|Gb\|_W^2. \quad (49)$$

PROOF. Full column rank gives $GA = I$. The error is $Gb + G\eta$, its covariance is $G\Sigma G^\top = M^{-1}$, and the cross term has expectation zero. Expanding the squared norm proves the result without a Gaussian assumption. $\square$

With no bias, minimizing this risk is weighted A-optimal design. D-optimality instead maximizes $\det M$ and generally optimizes a different objective. Under a correctly specified Gaussian model with parameter-independent covariance, M is the Fisher information. Full rank makes GLS well-defined, but unknown bounded bias can still prevent exact identification of θ .

For MT, fix the context distribution, clean states h_c , and accessible directions D_c . For mask a_j and scale $\varepsilon_j > 0$, set

$$\theta = \mathbb{E}_c[D_c^\top \nabla f_c(h_c)], \quad \kappa_j^2 = \mathbb{E}_c\|D_c a_j\|_2^2. \quad (50)$$

Under the lemma's activation-space Hessian bound L , the mean normalized response satisfies

$$F_j = \frac{\mathbb{E}_c[f_c(h_c + \varepsilon_j D_c a_j) - f_c(h_c)]}{\varepsilon_j} \quad (51)$$

$$= a_j^\top \theta + r_j, \quad |r_j| \leq \frac{L\varepsilon_j}{2} \kappa_j^2.$$

The bound follows by integrating the quadratic Taylor remainder along each segment and then averaging. If an additional systematic error is bounded by $\bar{\tau}_j/\varepsilon_j + \bar{v}_j$, the total envelope is

$$|b_j| \leq \beta_j := \frac{L\varepsilon_j}{2} \kappa_j^2 + \frac{\bar{\tau}_j}{\varepsilon_j} + \bar{v}_j. \quad (52)$$

The two kinds of error enter the model differently. Deterministic error, such as a systematic approximation in the response numerator, is part of the bias b_j and is bounded through $\bar{\tau}_j$. Centered sampling variation, including the covariance created when measurements share a random baseline, belongs in Σ . Each error is assigned to exactly one of the two terms. Lemma 1 bounds their magnitudes, so the zero-mean and covariance structure in (47) is an additional assumption.

COROLLARY 3 (BIAS-AWARE DESIGN CRITERION). *Let $B = \text{diag}(\beta_1, \dots, \beta_m)$ and let g_j denote column j of G . The risk of*

the physical response is at most

$$J_{\text{box}}(S) = \text{tr}(WM^{-1}) + \max_{s \in \{-1,1\}^m} s^\top BG^\top WGBs, \quad (53)$$

$$J_{\text{box}}(S) \leq J_{\text{sum}}(S) := \text{tr}(WM^{-1}) + \left(\sum_j \beta_j \sqrt{g_j^\top W g_j} \right)^2. \quad (54)$$

The first expression is the exact worst-case GLS risk over the rectangle $|b_j| \leq \beta_j$. For a scalar target $v^\top \theta$, so $W = vv^\top$, the two expressions coincide:

$$J_{\text{box}} = J_{\text{sum}} = v^\top M^{-1}v + \left(\sum_j \beta_j |v^\top g_j| \right)^2. \quad (55)$$

PROOF. The squared bias is a convex quadratic in b . Every point of the rectangle is a convex combination of its vertices Bs , so its maximum occurs at a vertex. The triangle inequality gives $\|W^{1/2}Gb\|_2 \leq \sum_j \beta_j \|W^{1/2}g_j\|_2$. For a scalar target, choose each sign to agree with $v^\top g_j$; this attains the bound and proves (55). $\square$

Minimizing J_{box} over a finite set of feasible designs therefore selects, among the declared GLS designs, the one with the smallest worst-case risk over the rectangle. This comparison holds the estimator family and error model fixed; it does not optimize over all estimators or physical response classes. For a scalar target, the criterion reduces to the closed-form sum in (55). For general W , evaluating J_{box} can require a search over the 2^m sign vectors. The cheaper bound J_{sum} can overstate the worst squared bias by up to a factor of m . That factor follows from Cauchy–Schwarz, since the average over vertices equals the sum of squared weighted-column norms; equal-norm orthogonal weighted columns attain it.

The rectangle can also be conservative. When probes lie on the same ray at different scales, their errors arise from the same curvature and need not attain their separate extremes jointly. Under a Hessian bound alone, however, the rectangle can be sharp as a supremum, as the signed example below shows. In every case, the curvature bound and covariance are inputs that must be known or estimated, and estimating them counts toward the measurement budget.

E.2 Why the smoothness assumption matters

Shared curvature can make the coordinatewise envelope conservative. Suppose the response in mask coordinates is exactly quadratic:

$$f(z) - f(0) = \theta^\top z + \frac{1}{2}z^\top H z, \quad H = H^\top, \quad \|H\|_{\text{op}} \leq \Lambda. \quad (56)$$

The constant Λ bounds curvature in mask coordinates and differs from the activation-space bound L : for a fixed direction matrix D , the pulled-back Hessian $D^\top H_{\text{act}} D$ has operator norm at most $L\|D\|_{\text{op}}^2$. Assuming a shared quadratic is

stronger than assuming a Hessian bound. Under that assumption, with no other systematic error and a fixed covariance, define

$$K_S = \frac{1}{2} \sum_j \varepsilon_j (v^\top g_j) a_j a_j^\top. \quad (57)$$

Then the exact worst-case scalar risk for the declared GLS estimator is

$$v^\top M^{-1}v + \Lambda^2 \|K_S\|_*^2, \quad (58)$$

where $\|K_S\|_*$ is the sum of absolute eigenvalues. To prove this, use $b_j = (\varepsilon_j/2)a_j^\top H a_j$ to write the scalar bias as $\text{tr}(H K_S)$. If $K_S = U \text{diag}(\lambda_i) U^\top$, its absolute maximum is $\Lambda \sum_i |\lambda_i|$, attained by $H = \Lambda U \text{diag}(\text{sign } \lambda_i) U^\top$. Unlike the rectangle, this expression allows cancellation before taking absolute values.

For a general scalar f with $|f''| \leq L$, however, the central difference has worst-case bias $Lh/2$, the same as the forward difference. Separate Taylor bounds at h and $-h$ give the upper bound. To approach it, choose $f'_\delta(t) = L \tanh(t/\delta)$, $f_\delta(0) = 0$, and $f'_\delta(0) = \theta$. Direct integration gives

$$\frac{f_\delta(h) - f_\delta(-h)}{2h} - \theta = \frac{L\delta}{h} \int_0^h \log \cosh(t/\delta) dt \rightarrow \frac{Lh}{2}. \quad (59)$$

Indeed, the bias lies between $L(h/2 - \delta \log 2)$ and $Lh/2$ because $x - \log 2 \leq \log \cosh x \leq x$ for $x \geq 0$. With $L = h = 1$ and $\delta = 0.001$, the bias is approximately 0.4993073. The function is infinitely differentiable; smoothness without a quantitative higher-order bound does not guarantee cancellation.

Central differences improve on this worst-case rate under a stronger smoothness assumption. If the activation-space Hessian is Lipschitz with constant L_3 on the signed segments, the standard remainder bound gives

$$\left| \frac{f(h_0 + \varepsilon u) - f(h_0 - \varepsilon u)}{2\varepsilon} - \nabla f(h_0)^\top u \right| \leq \frac{L_3 \varepsilon^2}{6} \|u\|_2^3. \quad (60)$$

The second-order terms of the two Taylor expansions cancel, and each third-order remainder is at most $L_3 \varepsilon^3 \|u\|_2^3/6$. Subtracting the expansions and dividing by 2ε gives the bound, which a one-dimensional cubic attains. A shared quadratic therefore gives zero bias, while a Lipschitz Hessian gives bias of order ε^2 [31]. The scheme needs both signs of the intervention to be feasible, which may fail for patching or ablation, and both evaluations count toward its cost. Lemma 1 needs only the Hessian bound; the higher-order rate needs the additional smoothness stated here, which the clipped HMM readout of Sec. 5.1 lacks on paths that cross its clipping boundaries.

E.3 An equal-budget comparison

Consider $f(t) = \theta t + (q/2)t^2$ with $|q| \leq \Lambda$, a known baseline $f(0) = 0$, and a budget of four evaluations, each with independent noise variance σ^2 . If all four are taken at the same

positive step h , the slope estimate has variance $\sigma^2/(4h^2)$ and worst-case bias $\Lambda h/2$, so its worst-case risk is $R(h) = \sigma^2/(4h^2) + \Lambda^2 h^2/4$. At $\Lambda = 1$, $\sigma = 0.1$, and $h \in [0.1, 1]$, information-only selection chooses $h = 1$ with risk 0.2525, while minimizing R gives $h = \sqrt{0.1}$ and risk 0.05. Two evaluations at each sign instead cancel the shared quadratic bias and give risk 0.0025 at $h = 1$. The symmetric control therefore outperforms the best one-sided design here; the cancellation requires the shared quadratic, not merely a Hessian bound. If the baseline evaluation were itself noisy, it would add cost and covariance terms to each design.

E.4 Finite-response calibration has a different target

The HMM calibration in Sec. 5.2 estimates a gain for finite responses, not the local derivative. Fix its scale and evaluation distribution Q before choosing measurements. Let $p(a)$ be the known scalar prediction, $F(a)$ the mean finite response, and assume finite second moments and $\mathbb{E}_Q p^2 > 0$. Define

$$g_Q = \frac{\mathbb{E}_Q[pF]}{\mathbb{E}_Q[p^2]}, \quad e(a) = F(a) - g_Q p(a), \quad \mathbb{E}_Q[pe] = 0. \quad (61)$$

For a selected set S , let $I = p_S^\top \Sigma_S^{-1} p_S > 0$ and $\hat{g} = I^{-1} p_S^\top \Sigma_S^{-1} y_S$. Proposition 4 and the projection orthogonality give

$$\mathbb{E}(\hat{g} - g_Q)^2 = I^{-1} + (I^{-1} p_S^\top \Sigma_S^{-1} e_S)^2, \quad (62)$$

$$\mathbb{E}_\eta \mathbb{E}_Q[(p\hat{g} - F)^2] = \mathbb{E}_Q[p^2] \mathbb{E}(\hat{g} - g_Q)^2 + \mathbb{E}_Q[e^2]. \quad (63)$$

The cross term vanishes because $\mathbb{E}_Q[pe] = 0$ and Q is fixed independently of the fitting noise. For a gain target other than g_Q , the cross term generally does not vanish.

COROLLARY 4 (RAW-RESPONSE ENVELOPE FOR GAIN CALIBRATION). *If $F = p + r$, $|r(a)| \leq u(a)$, and $\mathbb{E}_Q[|p|u] < \infty$, set $\alpha_Q = \mathbb{E}_Q[|p|u]/\mathbb{E}_Q[p^2]$. Then*

$$|g_Q - 1| \leq \alpha_Q, \quad |e(a)| \leq u(a) + \alpha_Q |p(a)|. \quad (64)$$

PROOF. Substitution in (61) gives $g_Q - 1 = \mathbb{E}_Q[pr]/\mathbb{E}_Q[p^2]$. The triangle inequality bounds this by α_Q . Apply it again to $e = r - (g_Q - 1)p$. $\square$

For an exact raw attribution prediction $p(a) = \epsilon a^\top \theta$, Taylor's theorem gives

$$u(a) = \frac{L\epsilon^2}{2} \mathbb{E}_c \|D_c a\|_2^2. \quad (65)$$

The bound on this unnormalized response scales as ϵ^2 ; if the attribution map is itself approximate, its error must be added. Applying the corollary requires a curvature bound and the Q -moments of p , and any finite-response measurements used to obtain them count toward the budget. When L is unknown or very conservative, the criterion becomes expensive to apply or too loose to guide the design.

The residual envelope need not be tight even within a constant factor. If $F = 2p$ and $u = |p|$, then $g_Q = 2$ and $e = 0$, but the envelope is $2|p|$. With independent equal-variance noise, selected predictors of magnitude P and a constant raw envelope U give the scalar bound

$$\frac{\sigma^2}{mP^2} + \left(\frac{U}{P} + \alpha_Q \right)^2. \quad (66)$$

For fixed U and α_Q , both terms decrease with P . Thus normalized masks and a uniform envelope do not establish that large attribution responses are poor calibration choices. Showing that saturation makes large responses poor calibration choices would require additional structure in the residual or direct empirical evidence.

Scope of implementation. The analysis extends to other fixed linear estimators. For an estimator G_0 , the variance becomes $\text{tr}(WG_0 \Sigma G_0^\top)$ and the squared bias becomes $\|(G_0 A - I)\theta + G_0 b\|_W^2$; for ridge regression, this includes the shrinkage bias that a direct application of the GLS result would omit. Applying the analysis after earlier data have been used to choose the design requires the noise assumptions to hold conditionally on that choice. Every pilot, covariance, and curvature measurement also counts toward the cost. Because the argument fixes the design before the fitting measurements, it gives no guarantee for sequential selection. The results so far are analytic; the retained-data comparison below shows how information and prediction error can diverge.

A two-stage acquisition workflow. We outline how the criterion could be applied to a small calibration model. The target, intervention scale, and candidate family are fixed first, and a separate validation budget is reserved. A pilot stage of k_1 uniformly drawn probes measures the finite-response mismatch, while the design matrix shows its conditioning. The pilot reveals where the local approximation fails but does not by itself bound curvature at unmeasured interventions.

When justified bias envelopes and a specified covariance model are available, a refinement stage chooses a full-rank design of k_2 probes by minimizing J_{box} or its tractable upper bound J_{sum} over feasible designs. The reported selection method distinguishes an exact minimum from an approximation. If pilot estimates replace established envelopes or covariance assumptions, the rule is a heuristic whose sensitivity needs to be assessed. The fixed-design analysis applies conditionally on the pilot when the refinement design is then fixed and fresh fitting responses satisfy its noise assumptions.

Selecting k_2 distinct probes from N candidates allows $\binom{N}{k_2}$ subsets before constraints are applied. Using J_{sum} avoids the inner sign enumeration but leaves this outer search. Greedy additions from a full-rank seed or rank-preserving probe exchanges avoid exhaustive search, although neither has an

optimality guarantee here. Selection runtime is therefore reported separately from measurement cost.

The refinement responses fit the model, and the reserved validation measurements assess its predictions at the intended scale. A cost-matched comparison with uniform selection includes pilot, refinement, and validation work, together with any measurements used to estimate covariance or bias. We propose this workflow but have not evaluated it; the replay below tests only fixed selection rules.

E.5 Calibration-probe selection on retained HMM measurements

Scale	Selection	Squared bias	Subset variance	Total MSE
5	Largest-first	0.12699	0.00000	0.27823
5	Random	0.00807	0.04454	0.20385
5	Spread	0.07358	0.00000	0.22483
8	Largest-first	0.17359	0.00000	0.47405
8	Random	0.02214	0.08331	0.40591
8	Spread	0.09887	0.00000	0.39933

Table 6: Eight-probe calibration on retained HMM measurements. Entries average the cellwise terms in Eq. (67). Total MSE also includes the best-gain error L_Q^* , averaging 0.15125 at $\varepsilon = 5$ and 0.30046 at $\varepsilon = 8$. Largest-first and spread are deterministic, so their subset variance within each cell is zero; their results still vary across cells.

Under a scalar linear model with independent, zero-mean, equal-variance errors, choosing the largest absolute predictions maximizes information and is both A- and D-optimal. Finite responses need not follow that error model. We compare this rule with uniform random selection and a simple alternative that spreads probes across prediction magnitudes, using the retained HMM measurements from Sec. 5.2.

All three rules see only candidate identifiers and attribution predictions. They select from the same 96-mask pool, fit the same no-intercept gain, and are scored on the same 32 held-out masks. The spread rule chooses unused masks nearest to equally spaced targets over the range of absolute predictions. Random selection uses 50 uniform draws in each cell; the same draws are used at both scales, and smaller subsets are nested within larger ones. The primary comparison, fixed before the replay, is mean held-out MSE at $\varepsilon = 8$ with eight probes, averaged equally across the nine cells. A probe is one mask response averaged over 256 sequences; held-out evaluation uses 32 additional mask responses. This replay makes no new model calls.

Selection changes calibration error. At eight probes, largest-first has 16.8% higher mean MSE than random at $\varepsilon = 8$ and 36.5% higher MSE at $\varepsilon = 5$ (Table 6). It is worse in five of nine cells at $\varepsilon = 8$ and eight of nine at $\varepsilon = 5$. Spread improves on largest-first at both scales, but is 10.3% worse than random at $\varepsilon = 5$ and only 1.6% better at $\varepsilon = 8$. The latter MSE difference, 0.00659, is smaller than the conditional Monte Carlo standard error of the random mean, 0.00789. With four probes, largest-first beats random selection only at $\varepsilon = 8$; with sixteen it loses at both scales, and with the full pool all three rules select the same masks. Because this comparison redraws the random subsets, its eight-probe means ($R^2 = 0.953$ at $\varepsilon = 5$ and 0.949 at $\varepsilon = 8$) differ slightly from those in Sec. 5.2 (0.951 and 0.948), which use the original draws.

An exact bias–variance breakdown. To explain these errors after the comparison, let F denote the retained response and Q the uniform distribution over one cell’s held-out masks. Let g_Q be its least-squares gain and $w = \mathbb{E}_Q[p^2]$. Writing $L_Q(g) = \mathbb{E}_Q[(gp - F)^2]$ and $L_Q^* = L_Q(g_Q)$, projection orthogonality gives the exact identity

$$\mathbb{E}_S L_Q(\widehat{g}_S) = L_Q^* + w(\mathbb{E}_S \widehat{g}_S - g_Q)^2 + w \text{Var}_S(\widehat{g}_S). \quad (67)$$

Here S ranges over the saved subsets, with equal weight on each of the 50 random draws. The variance measures variation across these subsets, not fresh measurement noise. The held-out gain is used only for this post-hoc diagnosis; it is never available to a selector.

At $\varepsilon = 8$, largest-first fits a mean gain of 0.642, compared with 0.742 for the held-out best gain; at $\varepsilon = 5$ the corresponding values are 0.749 and 0.886. Random selection has less squared bias but nonzero subset variance. Its squared-bias term accounts for about 21% of its excess error above the best-gain error at $\varepsilon = 8$. The comparison shows how calibration bias can outweigh a variance-based selection advantage. The decomposition attributes the excess error to bias in the fitted gain. Identifying the mechanism behind that bias, such as saturation, and testing the bias-aware criterion prospectively would require further measurements of curvature and covariance. The comparison covers nine combinations of evaluation batch and mask design from a single trained model.